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NERD

Nutrition Examination
Research Digest

THE MOST INTERESTING FITNESS ARTICLES FROM 2020



Letter from the editor (cont)

Thanks for buying the Examine Fitness Guide: A blueprint for optimizing nutrition and supplementation. As a special thank-you bonus, we're including this set of fitness-related articles from another Examine product: the [Nutrition Examination Research Digest \(NERD\)](#).

These articles provide a different perspective than the Fitness Guide. The guide's goal is to give you an overview of the best evidence on all six facets of fitness, from exercise and nutrition to understanding hormone regulation and joint health. You'll explore different dietary approaches that are both sustainable and compatible with your lifestyle, as well as learn about how the different aspects of fitness are interconnected. Then, we review the most effective strategies for optimizing each of those facets. The guide's aim is to equip you with the evidence-based tools you need to achieve your personal fitness goals through smart, intentional action. In short, you'll get what's on the label: a holistic, no-hype guide to fitness!

NERD is not dedicated to a single topic. Instead, the editors dive deep into the latest research coming out of the world of health and nutrition. We scour the literature from the past few months and choose articles based on a variety of factors: Are the contents making the news? Does the research make an important contribution to nutrition science? And most importantly: Does the study include information that our readers will find useful and interesting?

Over the past several years, interest in several methodologies for improving health and fitness — from high-intensity interval training to intermittent fasting and the ketogenic diet — has grown massively among both health enthusiasts and scientists. For example, a quick search of PubMed reveals over 3,000 scientific articles on the ketogenic diet, with around half of them published in the past five years alone. Not only is there more information than ever before, but people are increasingly spending more money on products purported to improve their health. The dietary supplements industry in the U.S. captured over \$35 billion of market share in 2020 and continues to grow by millions of dollars per year. With so much information and so many options to sift through, NERD has kept pace by covering some of the most interesting health-related studies each month and distilling them into digestible, actionable take-aways for readers.

The present collection includes eight articles that we published in 2020, investigating the most cutting edge lines of questioning in supplementation and exercise recovery research, as well as an exclusive interview with a leading researcher in the field of sports nutrition. We recommend you take some time to check them out because we really think they serve as a great complement to the Fitness Guide.

Letter from the editor (cont)

That's because NERD covers some domain-specific nitty-gritty details relevant to each cutting-edge study we review. For example, you'll learn about the motivation behind each study: In "Supercharging muscle protein synthesis through essential amino acids," we start by discussing the mechanism powering muscle protein synthesis and its importance for health and fitness outcomes. As you read, you'll pick up tips about how to interpret studies on your own by reviewing the methods the researchers used, their reasoning for their choices, and our own critiques of those methods.

In short, this collection provides a more thorough understanding of the research at hand and helps you both with your fitness toolset, and your future adventures in the world of health and nutrition research.

I hope you enjoy the journey!

A handwritten signature in black ink, appearing to read "Gregory Lopez". The signature is fluid and cursive, with the first name "Gregory" and last name "Lopez" clearly distinguishable.

Gregory Lopez, Editor-in-Chief

Table of Contents

- The relationship between vitamin C supplementation and exercise remains unclear

Vitamin C supplementation could reduce post-exercise inflammation and oxidative stress, which in turn could boost recovery. But more evidence is needed to say whether or not this is the case.

- Adding protein to carbohydrates for better endurance performance

Supplementing protein and carbs boosts endurance more than carbs alone, but it may just come down to calories.

- D-ribose for diminishing DOMS

D-ribose drops DOMS, but doesn't pack a punch performance-wise.

- Interview: Jeff Rothschild, RD, CSSD, PhD(c)

What should we eat before exercise? To find out, we picked the lead author's brain of a recent review that answers that exact question!

- Deep Dive: Will nitrates improve your training performance? It depends!

Nitrates provide a small but significant performance boost, but dose, timing, and your baseline aerobic fitness matters.

- Can sweet and salty mouth rinses provide a taste of victory?

This study provides a pinch of evidence that mouth rinses could affect performance, but more evidence is needed to sweeten up confidence in this finding.

- Not just for making cakes: baking soda may improve exercise performance

Bicarb seems to boost muscular endurance a bit but may not affect strength.

- Supercharging muscle protein synthesis through essential amino acids

Adding essential amino acids to whey may boost muscle protein synthesis, but the dose matters!

The relationship between vitamin C supplementation and exercise remains unclear

Tags: [Vitamin C](#), [Exercise](#)

 **Study under review:** [Effects of vitamin C on oxidative stress, inflammation, muscle soreness, and strength following acute exercise: meta-analyses of randomized clinical trials.](#)

Quick Takes

- What question this study addressed: *How does vitamin C supplementation affect markers of oxidative stress, inflammation, and muscle soreness and strength after acute exercise?
- **How the question was answered:** Through meta-analysis of randomized controlled trials.
- **Who was studied:** Mostly healthy, active men who weren't athletes, although some athletes and women were involved in a handful of the trials.
- **What the intervention was:** 400 milligrams to 3 grams of vitamin C, ranging from one day to a month of supplementation, followed by mostly aerobic exercise.
- **What the main takeaway is:** Vitamin C supplementation had

no clear effect on any measured parameter shortly after the exercise session, except for one controversial measure of oxidative stress (lipid peroxidation via TBARS) and one measure of inflammation (IL-6).

- **Caveats:** There were a wide variety of trials measuring many outcomes with small numbers of participants, suggesting that there's a lot more room for more and better quality evidence.

What was the question?

Does vitamin C supplementation affect oxidative stress, inflammation, muscle soreness, or strength following a single bout of aerobic or anaerobic exercise?

Why was the question worth asking?

The beneficial health effects observed after performing regular exercise are due to the adaptations that occur in several systems in the body, like the cardiovascular and musculoskeletal systems. The specific types of adaptations depend on the exercise modality. In skeletal muscle, exercise can increase inflammatory processes and oxidative stress acutely, which in turn trigger compensatory coping mechanisms. For instance, exercise has been shown^[1] to increase the production of reactive oxygen and nitrogen species, which serve as signaling molecules and increase the endogenous antioxidant capacity in the long term. The acute response to exercise, while important^[2] for long term adaptations, is also accompanied by reduced function^[3] of the muscle (i.e. decline in strength) and muscle pain or soreness caused by muscle damage. This is inconvenient for both regular people who want to exercise, as it can prevent people from continuing to exercise due to pain, or for athletes concerned about performance after exercise.

Vitamin C is a very popular supplement that can act as an antioxidant, and should, in theory, be able to reduce exercise-induced oxidative stress and inflammation. However, research on its effects on exercise-induced inflammation and oxidative stress, and related functional metrics, is still inconclusive.

Acutely, exercise increases inflammation and induces oxidative stress, which can cause a reduction in muscle strength and increase muscle soreness. This might be inconvenient for people who want to exercise frequently or athletes who need to maintain high performance after an exercise bout. Vitamin C could help attenuate this response, but the results to date are inconclusive.

How was the question answered?

Researchers conducted a random-effects meta-analysis of randomized controlled trials (RCT) following the PRISMA guidelines and Cochrane Handbook. The trial was registered in [PROSPERO](#). The study included data from 18 RCTs, with a total of 313 participants with a median age of 24 years. Of the 313 participants, 72% were men and only 13% were athletes. The others were described as active and healthy participants by the authors. The dose of vitamin C that was administered ranged from 400 milligrams per day to 3 grams per day, consumed orally, and aerobic exercise (i.e. running or cycling) was the main exercise modality (15 RCTs). Most studies (N=12) used a supplementation scheme for more than a week. The longest supplementation protocol lasted for 28 days.

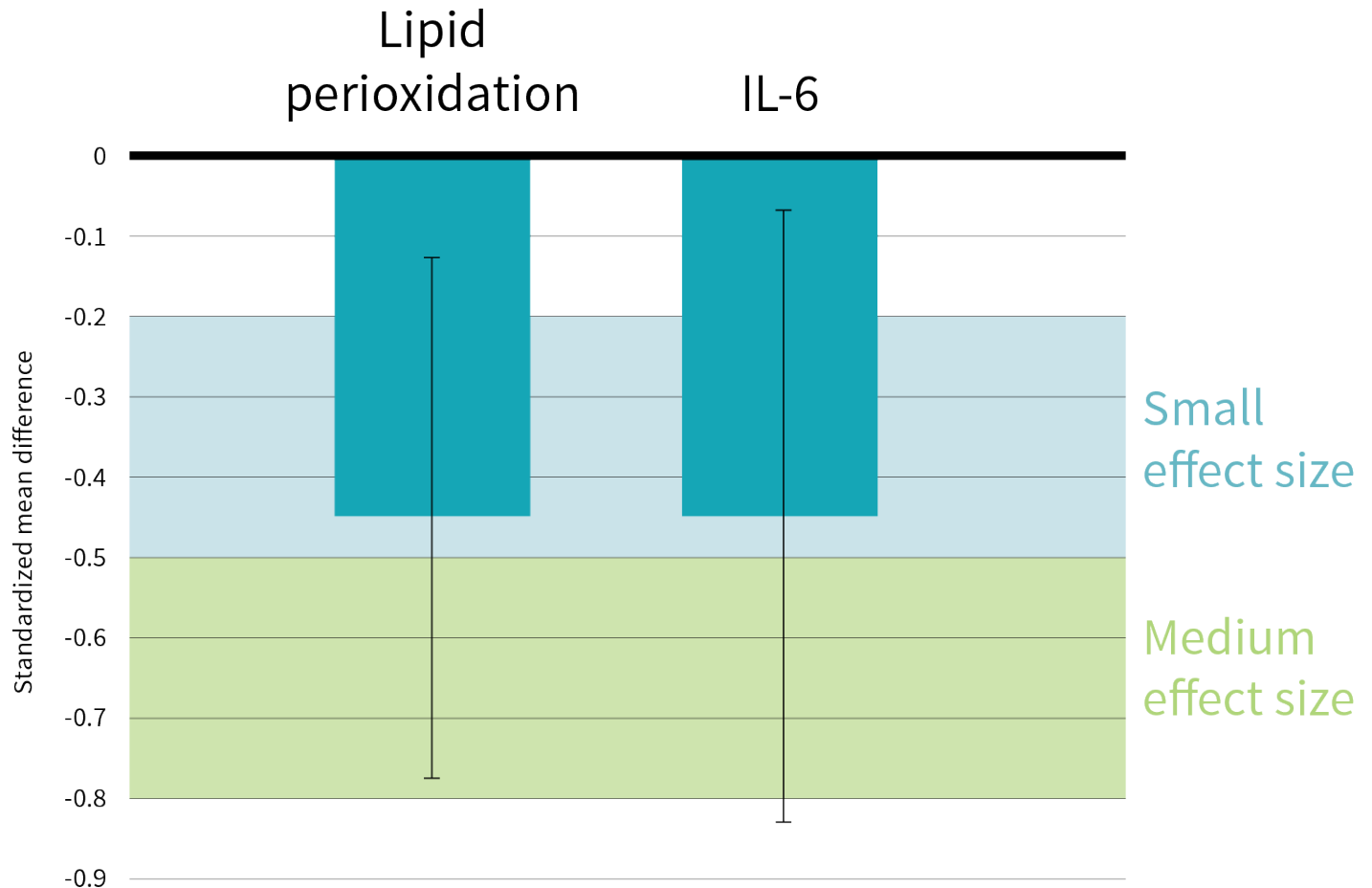
Lipid peroxidation (using MDA and TBARS) was included in the meta-analysis as a measure of oxidative stress, and inflammatory markers included interleukin 6 (IL-6) and C-reactive protein (CRP). Muscle damage was assessed using creatine kinase (CK) levels. Cortisol, muscle soreness and muscle strength, as a means to quantify performance detriments, were also included in the analysis.

The current meta-analysis included 313 healthy young participants (72%) supplementing with 400 milligrams per day to 3 grams per day of vitamin C. Most trials used aerobic exercise as the exercise modality and only 13% were conducted in athletes. The levels of lipid peroxidation, IL-6, CRP, CK, cortisol and muscle soreness and strength in response to vitamin C supplementation were analyzed.

What was the answer?

As shown in Figure 1, there was a statistically significant, but small in magnitude, reduction in lipid peroxidation acutely after exercise (up to two hours), but not over longer time periods. Subgroup analyses showed that the effect was significant when doses of vitamin C were higher than 500 milligrams, when exercise intensity was high and when lipid peroxidation was assessed using TBARS.

Figure 1: Effects of vitamin C on oxidative stress and inflammation markers up to two hours post-exercise (with 95% confidence intervals)



Similarly, vitamin C supplementation caused a small reduction in the levels of IL-6 only in the immediate two hours after exercise. On the other hand, vitamin C had no effect on CRP, muscle damage, muscle soreness, or muscle strength.

Overall, the risk of bias for most studies was unclear, and the quality of evidence (using GRADE) for lipid peroxidation and IL-6 up to two hours after exercise was moderate, although very low for lipid peroxidation after two hours.

Vitamin C supplementation lowered one measure of oxidative stress and one metric of inflammation two hours after exercise. No clear effects on muscle damage markers, muscle soreness, or muscle strength were observed.

How much should you trust the answer?

Overall, vitamin C showed no effect on most variables, and only a moderate effect on lipid peroxidation (assessed by TBARS) and IL-6, acutely after exercise. The very short lived effect of vitamin C supplementation agrees with its oral bioavailability: after ingestion, peak levels occur after about 40 minutes^[4]. This is why supplement timing might be of pivotal importance to elicit an effect. However, corresponding information about when the supplement was taken in the included studies was not consistently available. Another important consideration is that lipid peroxidation was only significantly reduced when assayed using TBARS, which has been shown to be flawed^[5] and is currently not recommended, especially in exercise contexts.

The main limitation of this meta-analysis is the very low number of RCTs that were available for analysis. For example, there were very few studies that analyzed other, more reliable markers of oxidative stress and thus couldn't be included in the analysis. Data for IL-6 also comes from only six RCTs, and the number of RCTs that tested muscle damage (N=9), soreness (N=7) and strength (N=5) was also very low. Moreover, the low number and heterogeneity of data precluded the analysis of differences by dosage, timing of supplementation, exercise modalities, or populations. The same is true for differences between athletes and recreationally active participants (given that the exercise sessions might be more demanding), or with resistance instead of endurance exercise, for which the acute inflammatory response appears to be important^[6] for hypertrophy and strength adaptations.

Vitamin C supplementation only had a minor effect lasting for a short period of time after an exercise session, which agrees with

its short bioavailability. The reduction in lipid peroxidation was only significant when it was determined using a technique which is not recommended due to methodological problems. Overall, the number of RCTs included in the meta-analysis was too low for some parameters of interest to be evaluated conclusively.

What's the take-home?

Vitamin C supplementation might moderately reduce oxidative stress and inflammation only acutely after exercise, but doesn't appear to affect measures of muscle damage, soreness, or strength. As the reduction in oxidative stress was observed only after researchers used a technique that is not recommended and because there was a low number of trials included in the meta-analysis, extreme caution is warranted when it comes to the interpretation of the results. This is all the more true for generalizations or predictions of long(er) term effects on adaptation. Future and larger studies are needed to properly evaluate the effects of vitamin C supplementation.

Exercise your brain by participating in the discussion around this study over at the [NERD Facebook forum](#)!

[^ Go back to table of contents](#)

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Adding protein to carbohydrates for better endurance performance

Tags: CHO, Protein, Endurance Training

Study under review: [The Effect of Ingesting Carbohydrate and Proteins on Athletic Performance: A Systematic Review and Meta-Analysis of Randomized Controlled Trials](#)

Quick Takes

- **What question this study addressed:** Does adding protein to carbohydrates improve athletic performance and reduce exhaustion?
- **How the question was answered:** Researchers addressed the question through a meta-analysis of randomized controlled trials.
- **Who was studied:** The participants were mostly active men who liked to cycle.
- **What the intervention was:** Researchers compared carbohydrate and carbohydrate-protein in a variety of ratios, with the most common being 4:1. Timing of the doses also varied between trials.
- **What the main takeaway is:** Protein-carbohydrate improved time trial performance and time to exhaustion more than carbohydrate alone.

- **Caveats:** The experimental and supplementation protocols varied widely between trials. Consistent performance benefits were observed only when treatments were standardized for carbohydrate intake, meaning when the CHO+PRO group consumed more calories.

What was the question?

Do carbohydrates combined with protein increase endurance performance compared to just carbohydrates in exercise time trials or time to exhaustion tests?

Why was the question worth asking?

The general recommendation for endurance athletes is to consume 5–12 grams of carbohydrates^[1] per kilogram of bodyweight per day. The suggested intake depends on the amount of exercise per day and proximity of an exercise event^[2], though recommendations vary among governing bodies. These recommendations come from a variety of sources that have been updated as new evidence is published, although they haven't changed much in the past decade. After general recommendations are met, athletes often look to gain marginal benefits by taking supplements or altering their nutrition timing^[3].

Carbohydrates are stored as glycogen in the liver and skeletal muscle. The rationale for consuming carbohydrates around the time of training is that they can be used more quickly for energy than fat through circulating blood glucose or glycolysis^[4]. There is also a rationale for consuming carbohydrates after training because glycogen depletion is associated with fatigue, reduced work capacity, and impaired cognitive function^[5], so athletes who compete in multiple events may need to refuel quickly. The exact mechanism of fatigue is unknown^[6] but may be due, in part, to the depletion of glycogen, which results in a reduction in the rate of ATP regeneration^[6].

Protein plays an important role in synthesizing new proteins and repairing muscle damage^[7]. Endurance athletes often focus less on protein consumption than athletes who resistance train because they don't need as much muscle mass. Recommendations for protein intake by endurance athletes range from 1.2–1.4 grams^[8] per kilogram of bodyweight per day. It is also

recommended that endurance athletes consume protein around their workout to enhance recovery and performance^[9].

The combination of protein and carbohydrates has been suggested to improve performance in various studies by reducing muscle damage^[10] or resynthesizing glycogen faster^[11]. The goal of this study was to determine if the combination of protein and carbohydrates could improve performance compared to carbohydrates alone and, if this effect occurs, when the optimal time is to consume such a supplement.

Carbohydrate recommendations for endurance athletes range from 5–12 grams per kilogram of bodyweight per day, and there may be ways to improve performance or recovery around training sessions by combining carbohydrate and protein intake.

How was the question answered?

This study was a systematic review and meta-analysis of randomized controlled trials on mid- or post-workout supplementation. Studies were included if they investigated the effect of the co-ingestion of carbohydrates and proteins (CHO-PRO) compared to only carbohydrates (CHO) in healthy adults. Studies with short-term (less than eight hours) and long-term (more than eight hours) recovery periods were included with data on time to exhaustion or time trials. A time trial is when athletes are given a certain distance to cover and their completion time is recorded. Time to exhaustion is when athletes are told to exercise at a certain rate, generally a percentage of VO_{2max} , for as long as possible.

Researchers completed sub-analyses to determine the effect ingestion timing and calorie content during time to exhaustion. Studies were also analyzed for risk of bias using the Cochrane Collaboration's risk-of-bias tool^[12], which includes details on the risk of bias from six domains: generation of the allocation sequence, concealment of the allocation sequence, blinding, incomplete outcome data, selective outcome reporting, and biases arising from randomization, deviations from the intervention, missing data, measurements, and selective reporting. This study was not preregistered.

The meta-analysis under review analyzed randomized controlled

trials that investigated carbohydrate ingestion or carbohydrate combined with protein ingestion during endurance exercise. The main outcomes were time to exhaustion or time-trial performance. Sub-analyses were conducted to explore short vs. long-term recovery periods, and caloric content. A risk-of-bias tool was used. The study was not preregistered.

What was the answer?

The authors identified 30 studies with 326 participants total, but only 11 people on average per study. The vast majority of the participants were male and all but one used a cross-over design. A cross-over design uses the same participants in both (CHO and CHO-PRO) groups. The strength of this design is that participants are their own controls, so confounding is lower if an appropriate wash-out period is used^[13]. Studies included data from participants who exercised on bikes or treadmills.

Figure 1: Overall results



CHO-PRO better than CHO
by 3.62 minutes



CHO-PRO better than CHO
by 1.5 minutes

This meta-analysis included 24 studies that compared the effect of CHO-PRO to CHO for time to exhaustion. As shown in Figure 1, the authors found a significant effect in favor of CHO-PRO, which equated to being able to exercise for 3.62 minutes longer. The heterogeneity in these studies was low, but the confidence intervals were large, indicating that the true size of the effect is unclear. The overall analysis on time trial performance favored the CHO-PRO group and indicated a 1.5 minute benefit. No sub-analysis was completed on time trials.

The authors then tried to determine if the timing of CHO-PRO ingestion played a role, so nine trials were analyzed that compared the effects of CHO-PRO to CHO during time to exhaustion tests. The authors found a significant effect in favor of CHO-PRO, which equated to exercising for 8.53 minutes longer. However, the 14 studies that analyzed the effect of CHO-PRO vs. CHO during the recovery period indicated no significant effect. This means that the timing of ingestion plays a role in the overall comparison, and taking CHO-PRO during exercise rather than after may provide more benefits.

The authors also completed a sub-analysis and found no effect of CHO-PRO vs. CHO on short-term recovery (no more than eight hours). They found a significant long-term effect (more than eight hours) equating to an additional 10.59 minutes in the time to exhaustion test. Two of the studies analyzed during the long-term sub-analysis used CHO or CHO-PRO, during and after the exercise bout, which could explain why the sub-analysis was significant since there were only three studies total. This, again, indicates that ingestion during exercise is key.

The authors also completed a sub-analysis of non-isocaloric CHO-PRO vs. CHO on time to

exhaustion using 12 studies. They found a significant effect in favor of CHO-PRO by 3.99 minutes. This helps tease out whether calories matter, rather than macro content alone. In another sub-analysis, the authors analyzed six carbohydrate matched studies and found a significant effect of CHO-PRO of 8.63 minutes. This implies that when carbohydrates were matched, protein played a major role in performance. The main takeaways from the subgroup analyses are depicted in Figure 2.

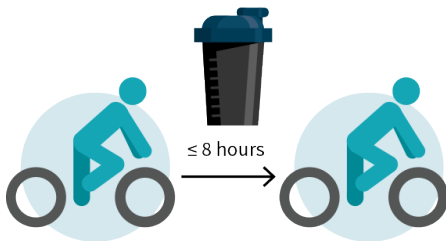
Figure 2: Subgroup analyses for time to exhaustion

Supplementation **during** exercise



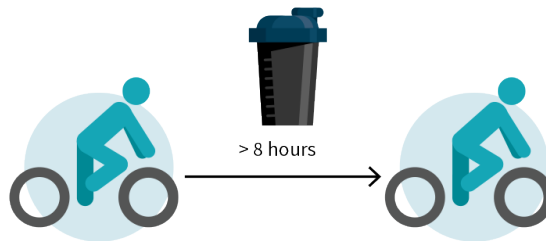
CHO-PRO better than CHO
by 8.53 minutes

Supplementation during **short-term** recovery



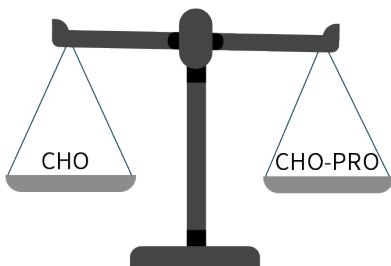
CHO-PRO $\approx$ CHO

Supplementation during **long-term** recovery



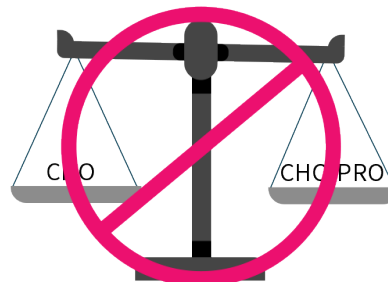
CHO-PRO better than CHO
by 10.59 minutes

Isocaloric supplementation



CHO-PRO $\approx$ CHO

Non-isocaloric supplementation



CHO-PRO better than CHO
by 3.99 minutes

Carbohydrate-matched supplementation

CHO = CHO
PRO

CHO-PRO better than CHO
by 8.63 minutes

The risk of bias assessment indicated that almost all studies had an unclear risk of random sequence generation, some had an unclear risk of selection bias or other bias, all studies had an unclear risk of detection bias (blinding of outcome), and most studies had a low risk of attrition or reporting bias. Overall, only six studies had a high risk of bias, each in only a single category, so these studies mostly were mostly well-designed.

CHO-PRO improves performance on time to exhaustion and in time trials compared to CHO alone. Sub-analysis indicates this is likely due to the consumption of CHO-PRO during exercise and may be due to the added calories of CHO-PRO, compared to CHO.

How much should you trust the answer?

The ingestion of carbohydrates and protein, together or independently, has been well studied to determine if it can improve endurance performance. Several studies indicate carbohydrates alone can improve performance directly by elevating blood glucose and glycogen resynthesis^[14]. Protein intake can help increase muscle size and repair muscle damage. Adding protein to carbohydrates potentially increases amino acid availability, increases insulin response, and potentially decreases cortisol levels^[15]. Thus, the authors hoped to harmonize the literature by using a meta-analysis to compare the effects of CHO-PRO vs. CHO on endurance performance.

The current study used time trials and time to exhaustion tests. Since there are large individual differences^[16] in the variability of time to exhaustion during submaximal exercise, using a meta-analysis to pool smaller studies together helps researchers understand the body of literature. Exhaustion trials are reproducible in trained cyclists^[17] and runners^[18] if they are exercising at the same relative intensity. However, there seem to be differences in the responses between trained and untrained participants^[19], in that trained participants adapt to higher intensity faster. The current study used data from participants with training backgrounds ranging from recreational athletes to elite cyclists. Furthermore, time to exhaustion tests allow for greater variability between studies because of the possible differences between fitness levels, which could allow for greater exercise time. For example, in Karp et al.^[20], participants cycled for 25–40 minutes depending on the group, while in Saunders et al.^[21], participants cycled for more than 100 minutes. The data was likely too small to conduct a sub-analysis for training status, so whether it plays a role is unclear.

The authors found that there was a benefit for CHO-PRO supplementation in non-isocaloric studies. This could be partially due to the increase in total calories when protein is consumed in addition to carbohydrates. A systematic review and meta-analysis^[15] by Stearns et al. found there was no significant improvement in cycling time to exhaustion with protein supplementation when compared to either an iso-carbohydrate or iso-caloric drink. In support of the current results, Stearns et al. also found that co-ingestion of protein with carbohydrate during exercise improved performance by about 9%, compared to carbohydrate alone. It would have been illuminating to review an overall isocaloric analysis in the current study.

The amount of muscle glycogen depletion determines, to a large extent, its rate of resynthesis^[22], so protocols that depleted more glycogen in the studies in this meta-analysis could influence the amount of glycogen replenished and thus the time trial or exhaustion outcome. The intensity of exercise in the studies analyzed ranged from 70–85% $\text{VO}_{2\text{max}}$ and the time trials ranged from 5–60 kilometers during the follow-up test. To give some context, a depletion protocol of 70% $\text{VO}_{2\text{max}}$ for about 50 minutes caused a 50% decline in muscle glycogen.

The rate of glycogen resynthesis begins after exercise and is highest during approximately the first six hours of recovery^[23], which is why the authors analyzed early (no more than eight hours) and late recovery (more than eight hours). Recommendations for post-exercise carbohydrate consumption are to consume 0.6–1.0 grams per kilogram^[9] of bodyweight within 30 minutes of finishing and every two hours for the next six hours^[9]. However, the relative carbohydrate intake of the participants was not provided. In cases of suboptimal post-exercise carbohydrate consumption, the addition of protein can improve glycogen replenishment and decrease symptoms of muscle damage^[9], so the relative amount could play a large role.

Even small differences in nutrition can have a significant effect on performance for athletes. For example, the difference between finishing first and second in the men's marathon at the 2016 Olympics in Rio was about 1%. Endurance athletes are often at risk of underperformance syndrome^[24], so recovery and performance must be adapted accordingly. If CHO-PRO helps tip the scale in favor of better performance, then it may be a worthwhile protocol for helping athletes train consistently while remaining healthy.

Carbohydrate plus protein consumption improves time to exhaustion and time trial performance in endurance athletes, but only in the short-term (no more than eight hours). This may be due to elevated glycogen resynthesis from the added protein.

What's the take-home?

There may be a marginal benefit from consuming carbohydrates in combination with protein, compared to carbohydrates alone, for improving endurance performance. This effect appears to occur when the nutrients are taken during or after exercise. This effect could be the result of the added calories from the additional protein, or the combination of carbohydrates, protein, and increased calories. The current study included a wide range of study protocols, and the participants had varied training statuses, which make the data difficult to fully interpret.

Discuss the results of this study with fellow NERDs over at our [Facebook group](#)!

[^ Go back to table of contents](#)

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D-ribose for diminishing DOMS

🔖 Tags: [Muscle Soreness](#), [D-Ribose](#)

📄 Study under review: [Effect of D-ribose supplementation on delayed onset muscle soreness induced by plyometric exercise in college students](#)

Quick Takes

- **What was the question?** How does D-ribose supplementation impact delayed onset muscle soreness (DOMS)?
- **How was it answered?** Researchers conducted a single-blind randomized controlled trial.
- **Who was studied?** Untrained healthy men participated in the study.
- **What was the intervention?** Participants took 15 grams of D-ribose or placebo one hour before, and several times after, a series of plyometric frog jumps.
- **What's the main takeaway?** Supplementation reduced DOMS by almost 50%, as measured by a subjective visual analog scale, up to 48 hours after exercise.
- **Any caveats?** While DOMS was reduced, D-ribose had no clear influence on performance. Also, the researchers weren't blinded to the condition, and the sample was small and limited to healthy untrained men.

What was the question?

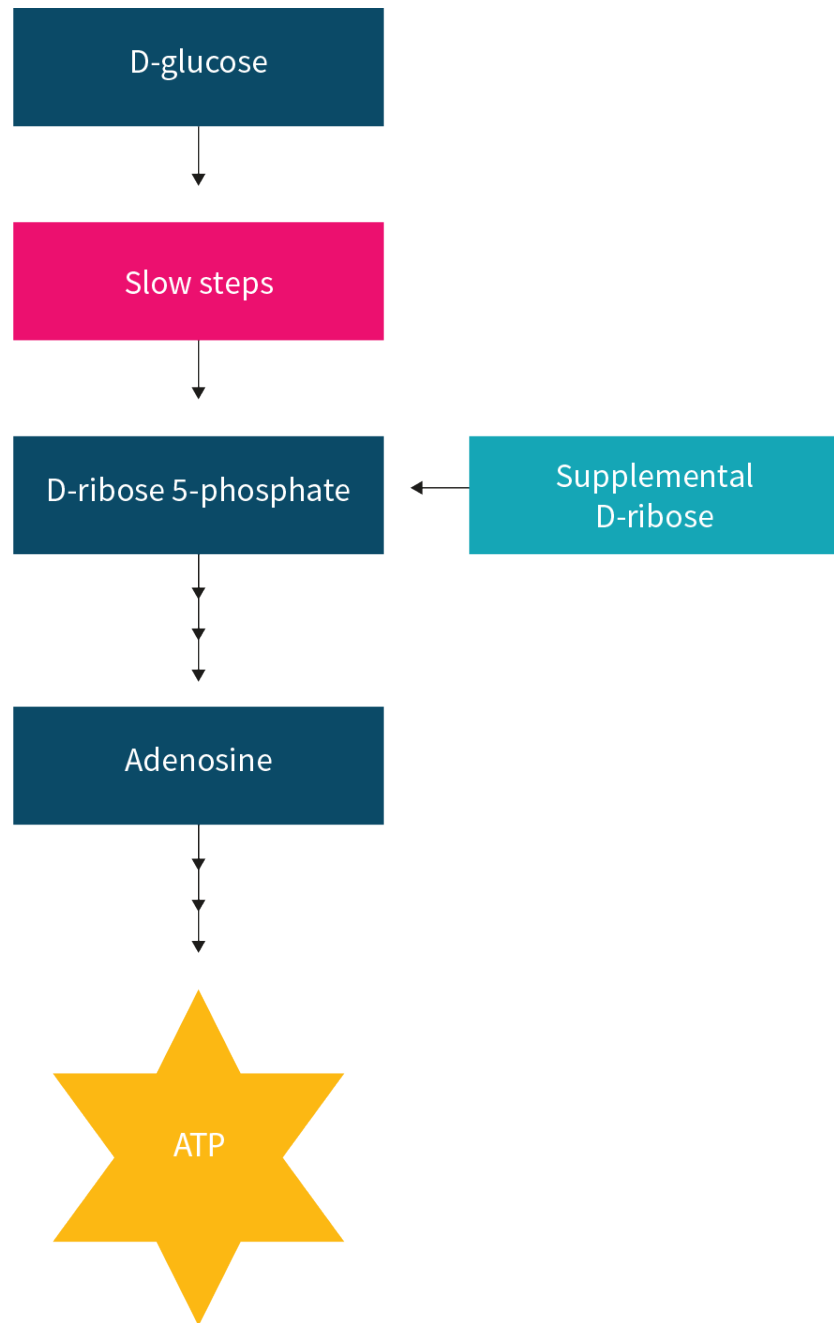
Can D-ribose supplementation reduce delayed onset muscle soreness?

Why was the question worth asking?

Delayed onset muscle soreness^[1] (DOMS) is muscle soreness caused by muscle tissue damage from exercise. It is characterized by muscle tenderness or soreness that develops over 24 hours after exercise, remains at its peak for 24–72 hours, and tapers off within seven days. This extended post-exercise recovery is one of the most common issues that hinders exercise training and performance. Possible mechanistic causes include excess mechanical damage^[2] that leads to inflammation, oxidative stress^[3] (from increased mitochondrial respiration^[4] as well as the tissue damage), and leakage of skeletal muscle enzymes^[5] and proteins through damaged cell membranes.

D-ribose^[6] is a naturally occurring monosaccharide that is a key component of DNA, RNA, and the main energy currency in the body: adenosine triphosphate (ATP). Supplemental D-ribose has been shown to enhance recovery^[7] of high-energy phosphates (such as ATP), presumably because it bypasses the slower enzymatic steps for its production from glucose and because D-ribose is a limiting factor^[8] for ATP resynthesis (Figure 1). Some studies have demonstrated improved^[9] exercise performance and/or recovery and reduced oxidative stress^[10] following D-ribose supplementation. The reduction in oxidative stress could mitigate some of the downstream biochemical effects that lead to DOMS. However, there have been conflicting results^[11] reported. It could be that training level^[12] is a factor. That is, well-trained people tend to recover more quickly, on average, so D-ribose wouldn't help them, which could explain some of the negative results.

Figure 1: How supplemental D-ribose could replenish ATP faster



Reference: Mahoney et al. Adv Biosci Clin Med. 2018 May.^[6]

The authors of the study under review aimed to investigate whether D-ribose supplementation after a bout of eccentric, DOMS-inducing exercise in young untrained men could reduce DOMS symptoms and associated biomarkers.

Evidence suggests that D-ribose can mitigate symptoms of delayed onset muscle soreness (DOMS) following exercise, but

some studies show conflicting results, which could depend on participant training level. The authors of the study set out to investigate the influence of D-ribose on DOMS in an untrained population.

How was the question answered?

The authors designed a single-blind (only participants did not know the treatment allocation) randomized controlled trial with 21 healthy untrained male participants to evaluate the impact of D-ribose supplementation on DOMS progression following a bout of plyometric exercise, in comparison to a placebo. They selected untrained people since they'd likely be more susceptible to DOMS, leaving more room for improvement through supplementation. The trial was not preregistered and primary or secondary outcomes were not distinguished.

Once enrolled, participants completed a plyometric exercise (an explosive movement that involves rapid deceleration, transition, and acceleration in the opposite direction) routine involving 7 sets of 20 consecutive frog hops (squat down, jump forward) with 90 seconds of rest between each set. Participants were then matched according to BMI and muscle soreness and randomly assigned to take either D-ribose (n=11, 15 grams mixed in 200 milliliters of drinking water) or the placebo (n=10, same dose and taste of a calorically equivalent sweetener as the treatment) after a second round of plyometric exercises that was completed two weeks after the first. Supplements were administered one hour before and 1, 6, 12, and 36 hours post-exercise.

Muscle soreness was assessed using a visual analog scale (0 = none, 10 = extreme) that participants were familiarized with before beginning the trial. Muscle strength was measured using a dynamometer that participants were also familiarized with before the trial. Blood samples were analyzed for creatine kinase, lactate dehydrogenase, and myoglobin as markers of muscle damage, and superoxide dismutase, total antioxidant capacity, and malondialdehyde as markers of oxidative stress. All measures were conducted at one hour before and 24 and 48 hours post-exercise.

Researchers conducted a single-blind randomized controlled trial, involving 21 healthy untrained men who supplemented with D-

ribose or an isocaloric placebo following a bout of plyometric exercise. Muscle soreness and strength, as well as markers of muscle damage and oxidative stress, were monitored up to 48 hours after exercise to measure the impact of the intervention on the progression of DOMS.

What was the answer?

D-ribose supplementation decreased muscle soreness by 49% at 24 and 48 hours after the exercise, compared to placebo. All included measures of muscle damage, and one measure of muscle oxidative stress (malondialdehyde) were also reduced. Strength data suggested an attenuation of DOMS-associated reductions in muscle strength following D-ribose intake, but results were not statistically significant between groups.

All participants completed both plyometric exercise sessions and reported an average muscle soreness of 5–6. No differences were found between groups in any measures at baseline.

After a bout of plyometric exercise, D-ribose supplementation reduced muscle soreness, damage, and oxidative stress in healthy untrained men. However, it didn't affect performance.

How much should you trust the answer?

It appears that the plyometric exercise was more than enough to induce DOMS. Previous studies that investigated the impact of plyometric exercise on performance, inflammation^[13], soreness^[14], and other health markers triggered a lower soreness VAS rating than the study under review. This is likely due to greater rest intervals and less exercise frequency.

D-ribose supplementation, based on its physiological function in the human body, should theoretically improve recovery. The body can either^[15] produce D-ribose from glucose, which is slow and depends on an enzyme that is often in short supply, or recycle ribose, assuming there is some to be recycled. Supplementation of D-ribose is a shortcut for both pathways, reducing the

need for the slow production process and providing a bulk supply to be used immediately. The presumed mechanism is that, with more ribose, the body can produce more nucleotides and ATP in response to damage and energy usage, respectively, that translates into faster recovery. However, this hasn't been explicitly tested in the context of DOMS, so this mechanism is speculative.

Overall, the underlying mechanisms are far from being well established. Given a presumed faster recovery that would allow for fewer muscle damage markers, lower inflammation, and increased ATP production, there may be some reason to believe that D-ribose can reduce oxidative stress. However, only one previous study (beyond the one under review) investigating oxidative stress^[10] induced by a bout of exercise had results that suggest D-ribose can help improve these changes. This previous study also had a very different treatment protocol. The concept still needs more research to flesh out a concrete answer, especially since mitochondria (the cell's ATP factory) create oxidative stress^[16] when producing ATP, either aerobically or anaerobically^[17], and if there's more ATP production because D-ribose is supplemented, more oxidative stress may be produced as well.

The decreases in markers of muscle damage and soreness observed in the study under review also seem like they would make sense with the physiological understanding of D-ribose, but there is a lack of data to back up these findings in this specific context. The study^[12] that only found an effect from D-ribose in untrained individuals reported improved ratings of perceived exertion and one marker of muscle health, creatine kinase. This supports the results of the study under review, but further studies evaluating D-ribose supplementation and its relationship with soreness, muscle damage, and oxidative stress are needed. Moreover, several markers^[18] exist for muscle damage and researchers are not exactly clear on the relationships between all these markers and exercise-induced muscle damage, so a standardized method needs to be established. Finally, studies on trained individuals need to be conducted to be able to conclude whether or not trained individuals would benefit from supplementation, and through which underlying mechanism.

Previous studies demonstrate conflicting results^{[19][20]} regarding changes in strength reductions from exercise following D-ribose supplementation. While performance could be regarded as the most important metric when evaluating a sports supplement, D-ribose has not demonstrated^[21] much promise for anaerobic activity^[20] performance, as shown in the examples in Figure 2. Many differences in outcomes across the literature may have to do with varying dosage, type of exercise, duration of treatment, and other factors. A systematic review and meta-analysis would be very helpful in order to shed some light on these details, once more studies are conducted.

Figure 2: Some instances in which D-ribose has not shown performance benefit

10 grams daily for 8 weeks
had no effect



on time trials for female rowers

10 grams for 5 days
had no effect



on metrics of anaerobic capacity
in resistance-trained males

2 three-gram doses before
and during cycling sprints
had no effect



on anaerobic
performance in male cyclists

32 grams over 36 hours
had no clear effect



on cycle sprinting performance
in formerly competitive,
recreationally active males

References: Dunne et al. Clin J Sport Med. 2006 Jan.^[22]|Kerksick et al. Int J Sport Nutr Exerc Metab. 2005 Dec.^[21]|Kreider et al. Int J Sport Nutr Exerc Metab. 2003 Mar.^[20]|Berardi et al. J Strength Cond Res. 2003 Feb.^[19]

There are no adverse effects reported from short-term D-ribose supplementation, but reports on long term use are lacking^[23].

The study under review was single-blinded, meaning that only the participants did not know which treatment they were receiving, while the researchers knew the group each participant was assigned to. Whether consciously or unconsciously, researchers who know which participant receives which treatment might carry a bias^[24] throughout data collection and analysis that ends up favoring a certain result. Since this was a pilot study (the first of its kind) with a small sample size, and there was little monitoring of compliance, and no dietary control, the generalizability of the reported results are limited.

While the positive results from the study under review regarding D-ribose supplementation for DOMS symptoms and associated markers in untrained men may make physiological sense, they are quite preliminary, so further studies are needed to confirm the findings and broaden their applicability.

What's the take-home?

After a bout of plyometric exercise by healthy untrained men, D-ribose supplementation was found to reduce muscle soreness, damage, and oxidative stress, but not the exercise-induced strength decline. This was a pilot study, so more controlled and careful research must be conducted to construct a data-driven understanding of the relationship between D-ribose and exercise recovery, rather than the current physiological speculation.

Are you going to start taking D-ribose in an attempt to shorten your exercise recovery? Discuss in our [NERD Facebook Forum!](#)

[^ Go back to table of contents](#)

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Interview: Jeff Rothschild, RD, CSSD, PhD(c)

Tags: [NERD Interview](#)



Jeff Rothschild is a Registered Dietitian and Board-Certified Specialist in Sports Dietetics. Jeff is currently doing his PhD research at Auckland University of Technology in the Sports Performance Research Institute New Zealand (SPRINZ) research lab, under the guidance of Dr. Dan Plews and Prof. Andrew Kilding. He is looking at the effects of pre-exercise nutrition on adaptations to endurance training, trying to better understand how what we eat before exercise affects the responses to exercise training.

Q. You recently co-authored [a review article](#) exploring what we should eat before exercise. There, you described how carbohydrate restriction between training sessions can be used to lower muscle glycogen and thus increase fat oxidation. From a practical

standpoint, if someone is interested in reducing fat, could it be as simple as doing cardio *after* resistance training? Or would longer intervals between the first and second training session be required?

Well, doing cardio after resistance training opens up the door to interference effects and the area of concurrent exercise training. But if someone's goal is simply to stay fit and healthy, then yes, doing cardio at the end of a resistance training workout would be a fine idea. It really depends on the person's goals, but it may be better to separate the sessions. One option to perhaps gain some additional benefit could be to do a resistance training session in the morning (i.e. before going to work), only consume protein afterwards, and then do a cardio session around midday before eating lunch. It's also important to note that burning more fat (compared with carbohydrate) during exercise doesn't necessarily mean it is the best way to reduce body fat.

Q. The review paper also mentions that AMPK, which inhibits ATP-producing consuming pathways and stimulates ATP-producing pathways, can be activated to varying extents by exercise intensity, training status, muscle glycogen, and nutrient availability. How quickly do we see this AMPK activation in response to training? Is this the type of thing someone needs to be concerned about within a particular training session?

This happens quickly (i.e., during exercise), but it is not something people need to think much about since the only way to measure it is with a muscle biopsy. With that said, it is safe to assume that undertaking exercise with reduced glycogen levels (by restricting carbs in between two workouts within a short duration) may be able to up-regulate this signaling inside our muscles.

Q. The review also mentions that, in general, being fasted or fed has a larger effect on performance than the size or timing of meals, and that fed exercise seems

to be better for performance. Is there any evidence that individuals who are accustomed to training in a fasted state have less of a dip in performance?

The key thing here is likely to be the exercise duration. During shorter-duration exercise (less than 60 minutes), a placebo/belief effect is more likely to occur, and it could be possible for people to 'convince' themselves that their performance will suffer. For longer duration exercise, glycogen depletion becomes more of the issue and so I don't think being accustomed to training fasted will help very much, presuming we're talking about high-intensity exercise performance. There may be some mental benefit for people who are more used to the feeling of being a bit hungry, but I'm not sure how much that would affect things. I'm not sure there is a study testing this exactly, but in my research there hasn't been any influence of habitual fasted training practices on performance in the fed vs. fasted state.

Q. Similarly, you note that trained athletes tend to perform better after consuming carbohydrates prior to training, and that most athletes tend to believe their performance is better when they've eaten beforehand. Has there been any research that's tested performance in people who don't believe it matters whether they eat or not?

In our [survey](#) of nearly 2,000 endurance athletes, about 50% disagreed with the statement "The quality of my workout is the same whether I eat or do not eat beforehand," while the other half were split between 'agree' and 'neutral'. So we might expect to see differences based on personal beliefs. But as noted above, even among people that didn't expect to perform well in the fasted state, we've observed similar performance in the fed or fasted state when the exercise duration was around one hour.

Q. What might mediate the improvement in time-to-fatigue seen in fasted athletes?

That was seen in [one study](#) that found better improvements in time-to-fatigue at 85% $\text{VO}_{2\text{max}}$ in

trained cyclists performing four weeks of sprint interval training in the fasted state compared to those that consumed carbohydrate before and during exercise. In this case, I suspect it might be more to do with carbohydrate blunting the training adaptations, as opposed to fasted training improving it to an additional degree. To clarify, there seems to be the potential for carbohydrates to negatively impact training adaptations when consumed in very high amounts (and I think the amount is the key thing). In that study, the sprint training consisted of four to seven 30-second sprints, performed three times per week for four weeks. So it was a very low volume of exercise. The carbohydrate group consumed a very large amount of carbs both before and during exercise (2.5 grams per kilogram of bodyweight before, and sipping on sports drink during, training sessions). So it's possible that the huge amount of carbs consumed before exercise (about 200 grams) could have blunted some intracellular signaling in response to the training and blunted some of the adaptations. However, I wouldn't expect to see that result if the carb group had consumed something more modest like 50–75 grams before exercise. There's also the confounding factor of consuming the carb drink during exercise, which can influence things on its own.

Q. Should carbohydrate periodization be done on a micro, meso, or macro scale? In other words, should athletes spend days, weeks, or months in high or low-carbohydrate training conditions?

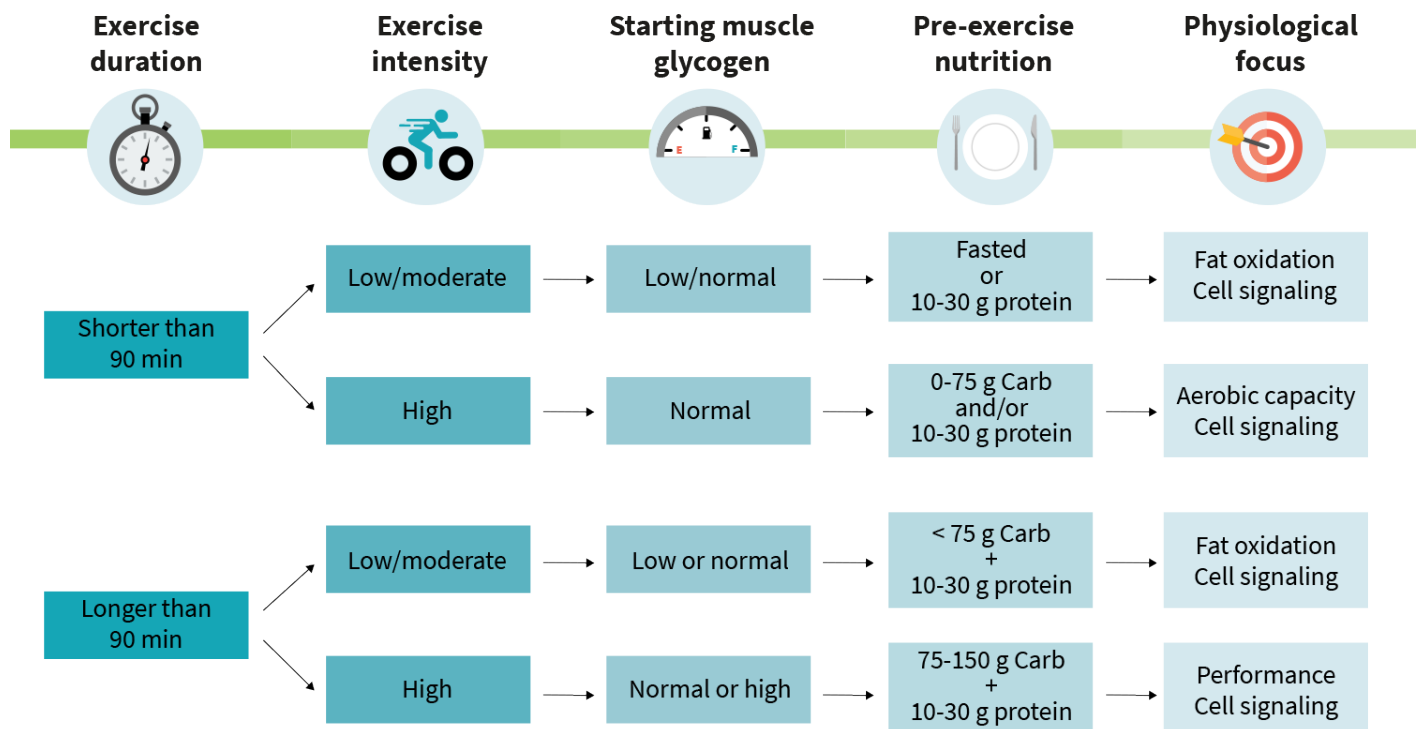
The prevailing sense is that it should vary on a micro scale—that is, day-to-day or even meal-to-meal. You can think of the last workout you had, and the next workout you have coming up, and load the fuel tank appropriately. With that said, it would also likely vary through the year as the training intensity and volume would also be changing through the year. So if you're eating to match your training on a day-to-day basis and most of your training is lower intensity and/or volume for a period of weeks or months, then your food should match closer to that.

Q. To wrap things up, can you run us through the practical takeaways of your research by answering the question the paper poses: What should we eat before exercise?

On one hand, for most people it probably doesn't matter that much. I feel odd saying that, but the training itself is going to matter so much more. But going beyond that, the duration and intensity of

the exercise session should be considered when deciding on the best pre-exercise nutrition choices. Before shorter duration exercise sessions that focus on lower intensity steady-state training, it may be beneficial to withhold carbohydrate, but there is little reason to think there are benefits to avoiding carbohydrate before high-intensity exercise. So assuming you're consuming less than approximately 75 grams of carbohydrate, food choices before high-intensity training sessions can be left to personal preference. For longer duration exercise (more than 90 minutes), there is little evidence to suggest fasted-state training offers any additional benefit, although this is still practiced by many endurance athletes. Ingesting less than approximately 75 grams of carbohydrate is unlikely to impair mitochondrial signaling adaptations from longer-duration, low-intensity exercise, while consuming 75–150 grams of carbohydrate prior to extended high-intensity exercise is suggested to increase endogenous fuel storage. Overall, I think there is more leeway than people often think. As mentioned in the above study, consuming 200 grams of carbohydrate before a short interval session is probably going too far and might be disadvantageous. But if you're consuming around 75 grams or less, I don't see much reason to think it will have undesirable effects.

Figure 1: What should we eat before we exercise? It depends!



Adapted from: Rothschild et al. Nutrients. 2020 Nov.^[1]

The views and opinions expressed by interviewees are their own, and do not necessarily reflect those of Examine.com.

[^ Go back to table of contents](#)

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Deep Dive: Will nitrates improve your training performance? It depends!

Tags: Nitrate, Supplementation, Exercise Performance, Aerobic

Study under review: [Ergogenic Effect of Nitrate Supplementation: A Systematic Review and Meta-analysis](#)

Quick Takes

- **What was the question?** How does nitrate supplementation impact performance?
- **How was it answered?** Researchers completed a meta-analysis of randomized controlled trials.
- **Who was studied?** Participants were healthy people, mostly 18–40 years old. Men were much more highly represented than women.
- **What was the intervention?** Participants supplemented nitrate 40–140 minutes before exercise, with doses ranging from roughly 1–30 mmol (about 60–1,800 milligrams). The studies used both acute dosing (one dose before exercise) and chronic dosing (multiple doses over several days). Outcomes included a wide variety of mostly aerobic metrics such as cycling or running performance.
- **What's the main takeaway?** Overall, nitrate

supplementation had a small but statistically significant effect on performance outcomes. The effects were strongest (but still very small to small) from doses of more than 5 mmol (310 milligrams) per day, after at least one day of supplementation, and when supplementation occurred 2–3.5 hours before training. These effects were only observed in participants with lower baseline aerobic fitness.

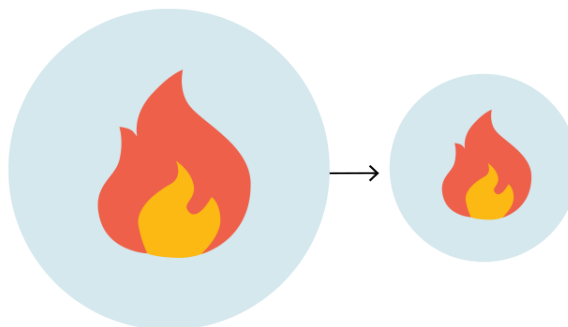
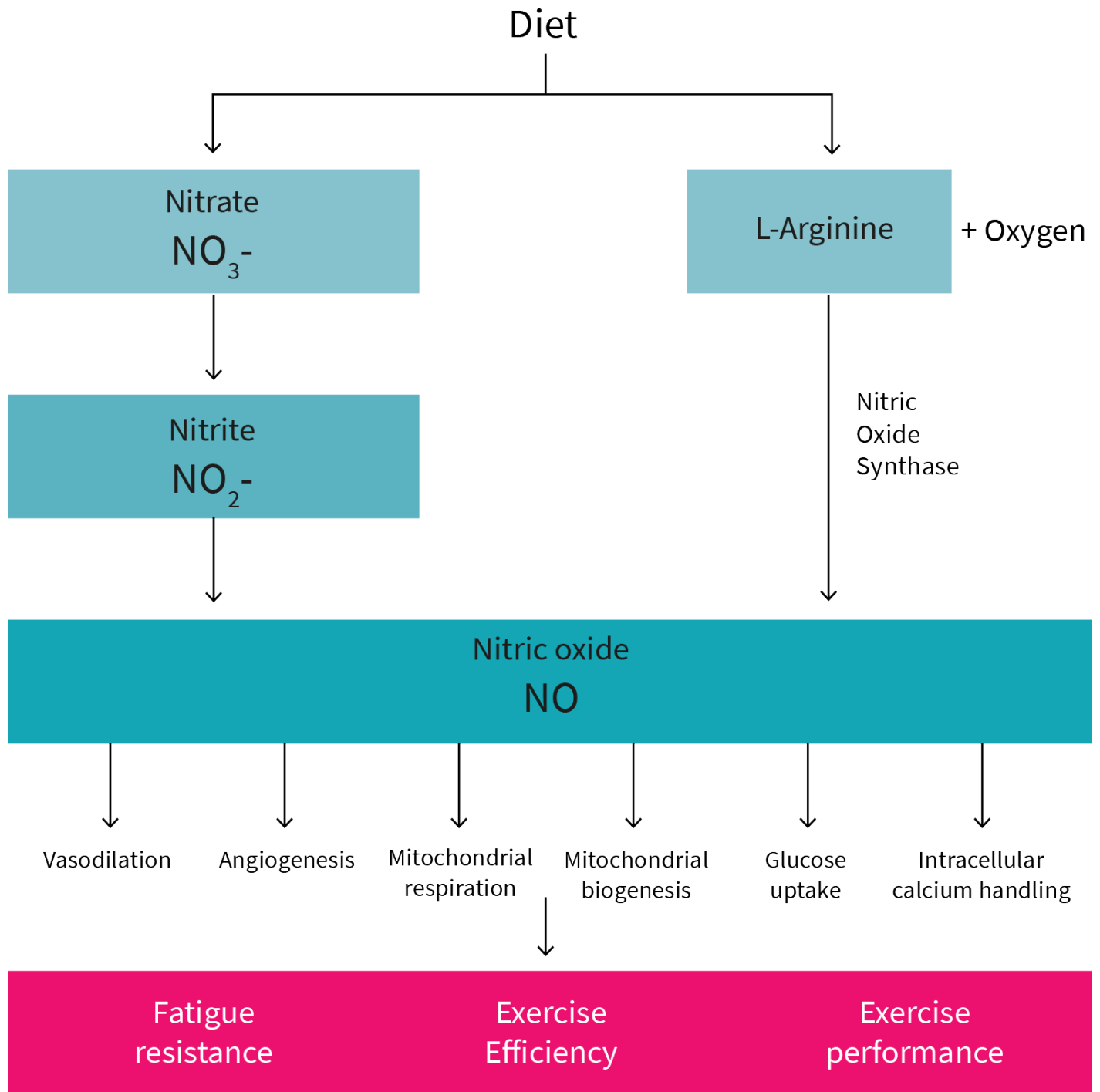
- **Any caveats?** Although the authors reported on publication bias, they didn't report on the quality of the included studies, making it difficult to determine how accurate these estimates were. Also, there was no statistically significant effect found on performance measures for knee extension and rowing. This could be due to a combination of small effect sizes and little data. Since these two exercises were very underrepresented, it's quite possible that more data could reveal a significant (but probably small) effect.

Introduction

Both athletes and recreationally active people commonly use dietary supplements in an attempt to improve their physical performance. One such supplement is nitrate (NO_3^-), a molecule which is produced in the body in small amounts, and which is obtainable through the diet mainly through the consumption of some vegetables^[1] (e.g., lettuce, beetroot, and spinach).

In the body, dietary nitrate is reduced to nitrite, and then to nitric oxide (NO) through what is called the nitrate-nitrite-nitric oxide pathway^[2], shown in Figure 1. NO, through its effects on vasodilation^[3], mitochondrial respiration^[4], calcium handling^[5], and glucose uptake^[6], may improve muscle function and reduce fatigue, thereby enhancing physical performance.

Figure 1: How dietary nitrate may ultimately boost performance



While NO_3^- supplementation should, theoretically, improve training performance, there is substantial variability in its ergogenic effects within and between trials. This variability could be due to several factors, including biological sex, aerobic fitness level, environmental oxygen concentrations, exercise parameters, and the dosage and timing of NO_3^- supplementation. The meta-analysis under review pooled the available trials together to examine the overall impact of nitrate supplementation on training performance, and is the first meta-analysis to have examined the potential influence of the aforementioned factors on NO_3^- ergogenicity.

Although nitrate supplementation is common in athletes and recreationally active people looking to improve their physical performance, there is substantial variability in the ergogenic effects of nitrate within and between trials. This variability could be due to several individual, environmental, exercise, and supplementation-related factors. The meta-analysis under review aimed to examine the potential influence of these factors.

What was studied?

This was a systematic review and meta-analysis that was designed to evaluate the efficacy of NO_3^- supplementation for improving training performance, and examined the potential influence of biological sex, aerobic fitness level, environmental oxygen concentrations, exercise parameters, and the dosage and timing of NO_3^- supplementation on NO_3^- efficacy. Eligible studies had to be randomized, single or double-blind, crossover, and placebo-controlled, and carried out in healthy people with a mean study age of 18–40 years.

Ultimately, 80 studies were included. Of these, two presented data from men and women separately, six presented data from different NO_3^- supplementation protocols separately, seven presented data from different exercises separately, and eight presented data from different environmental oxygen concentrations separately, resulting in 113 data sets. The studies involved a total of 1,135 participants (1,179 men and 156 women) with each study including 6–36 participants. The majority of studies ($n=52$) measured performance through time trials, with almost all of the others using other measures related to aerobic endurance. Only 13 trials looked at strength or power, and those examined effects on reduction of strength or power through fatigue.

To assess individual study quality, the meta-analysis used the [Cochrane Collaboration's tool](#) for assessing risk of bias in RCTs. Bias was assessed as high, low, or unclear over seven components: (i) sequence generation, (ii) allocation concealment, (iii) blinding of participants and personnel, (iv) blinding of outcome assessors, (v) incomplete outcome data, (vi) selective outcome reporting, and (vii) other sources of bias. However, the risk of bias assessment was not fully presented in the paper. Publication bias was assessed using a funnel plot, and no publication bias was detected.

The studies were pooled in a [random effects model](#), and the [standardized mean difference](#) (SMD) was used to determine the size of the effect, where an SMD of 0.15–0.2 was defined as very small, 0.2–0.5 as small, 0.5–0.8 as moderate, and more than 0.8 as large. The study was not preregistered.

This was a systematic review and meta-analysis evaluating the efficacy of NO_3^- supplementation for improving training performance, and examining the potential influence of several individual, environmental, exercise-related, and supplementation-related factors on NO_3^- efficacy. The meta-analysis included 80 crossover and placebo-controlled trials carried out in 1,135 healthy young adults (1,179 men and 156 women).

What were the findings?

In the overall analysis, NO_3^- supplementation improved performance, compared to placebo, with a very small effect size (SMD=0.174; CI: 0.12–0.23).

In the subgroup analyses by biological sex, NO_3^- supplementation outperformed placebo in men-only and mixed sex analyses, but the effect was smaller and not statistically significant the women-only analysis. In the subgroup analyses by aerobic fitness level, NO_3^- supplementation outperformed placebo in the first four aerobic fitness categories ($\text{VO}_{2\text{peak}}$ of less than 50, 50–54.9, 55–59.9, and 60–64.9 mL/kg/min) with an effect size hovering around very small to small (SMD of less than 0.25), but not in the highest aerobic fitness category ($\text{VO}_{2\text{peak}}$ of more than 65 mL/kg/min).

In the subgroup analyses by environmental oxygen concentrations, NO_3^- supplementation

outperformed placebo under both normoxia and hypoxia. In the subgroup analyses by exercise duration, NO_3^- supplementation outperformed placebo in all exercise duration categories (less than 300, 301–600, 601–999, and 1,000+ seconds), with the effect size being smaller in the longest-duration exercise category (1,000+ seconds or 17+ minutes). In the subgroup analyses by exercise type, NO_3^- supplementation outperformed placebo in cycling, running, and handgrip exercises, but not in knee extension and rowing.

In the subgroup analyses by NO_3^- dosage, supplementation with more than 5 mmol per day (310 milligrams per day) outperformed placebo, while the effect did not reach statistical significance with doses under 5 mmol per day. In the subgroup analyses by supplementation duration in days, NO_3^- supplementation outperformed placebo in all categories (i.e., 1, 2–6, and more than 7 days). Finally, in the subgroup analyses by supplementation timing, NO_3^- supplementation outperformed placebo when the supplement was taken 2–3.5 hours before training, but not when taken closer to training (i.e., less than 2 hours).

The subgroup analyses are a handful, so their main takeaways are summarized in Figure 2.

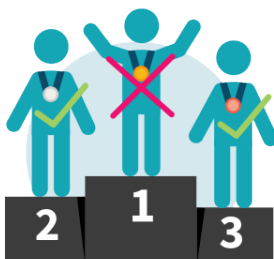
Figure 2: Main takeaways from the subgroup analyses

Nitrate supplementation tended to work best ...

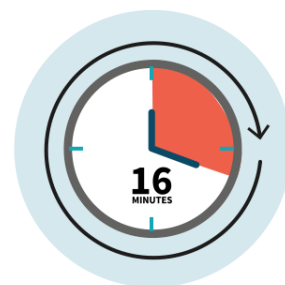
... in men
SMD $\approx$ 0.17



... in all but the highest
aerobic fitness category
SMD $\approx$ 0.16 - 0.24



... for exercise bouts under
~16 minutes SMD $\approx$ 0.22 vs
0.10 for longer bouts



... using doses over ~310mg/day
SMD $\approx$ 0.27



... when taken 2-3.5 hours
before training
SMD $\approx$ 0.16-0.24



NO_3^- supplementation slightly improved performance, as compared to placebo. In the subgroup analyses, the effects were significant in men-only and mixed sex trials, in participants with $\text{VO}_{2\text{peak}}$ of less than 64.9 mL/kg/min, in both normoxia and hypoxia, with all exercise durations, and with cycling, running, and handgrip exercises. The effects were also significant with NO_3^- dosages more than 5 mmol per day (310 milligrams per day), with at least one day of supplementation, and when NO_3^- was taken 2–3.5 hours before training.

The bigger picture

The study under review found a very small (about 3%) but statistically significant effect of NO_3^- supplementation for improving training performance. These results seem to support the mechanistic evidence relating to nitrate's potential ergogenic effects, and are in agreement with previous narrative^[7] reviews^[8] on the topic. It did, however, also confirm that there are substantial differences in the efficacy of NO_3^- ergogenicity between trials, with the subsequent subgroup analyses suggesting that the ergogenic effects of NO_3^- are more apparent in sedentary or recreationally trained adults (as opposed to highly trained athletes), and with higher-intensity, shorter-duration exercise (i.e., less than 16 minutes).

In the subgroup analyses according to sex, NO_3^- supplementation improved performance in men-only and mixed sex trials, but not in women-only trials. While the reasons for the differential ergogenic effects of NO_3^- in men and women are not clear, they may relate to anthropometric, metabolic, and physiological differences^[9] between the sexes that may affect the biological effects of NO_3^- in the body. Considering that the effect size of NO_3^- in women wasn't that much smaller, it may be that it didn't reach statistical significance just due to the low number of women-only trials included in the meta-analysis (n=6), which may have rendered the analysis underpowered to detect an effect.

Moreover, NO_3^- supplementation failed to improve performance in people with the highest aerobic fitness level (i.e., $\text{VO}_{2\text{peak}}$ of more than 65 mL/kg/min). This finding is in agreement with a 2013 meta-analysis^[10], which found that NO_3^- supplementation improved training performance in non-athletes ($\text{VO}_{2\text{peak}}$ of less than 65 mL/kg/min) but not in athletes ($\text{VO}_{2\text{peak}}$ of more than 65 mL/kg/min). One potential explanation for the blunted ergogenic effects of NO_3^- supplementation in athletes is that nitric oxide synthase activity is increased^[11] in highly trained individuals, which may mean that the nitrate-nitrite-nitric oxide pathway becomes less important for NO production. On the other hand, some research challenges this explanation, and, instead, suggests that highly trained individuals may have higher^[12] levels of plasma nitrite (due to eating a superior diet), which could result in a smaller response to NO_3^- supplementation.

In the subgroup analyses examining the influence of exercise duration, NO_3^- supplementation improved training performance in all exercise duration categories. However, the effect size was smaller when exercise lasted approximately more than 16 minutes. This makes sense from a physiological standpoint, as longer-duration, lower-intensity training results in a reduced^[7] requirement for production of NO through the nitrate-nitrite-nitric oxide pathway.

With regard to exercise type, NO_3^- supplementation led to small improvements in cycling and running performance, and non-significant improvements in knee extension and rowing performance. While, at first glance, these results would suggest that the ergogenic effects of NO_3^-

supplementation may depend on exercise type, it should be kept in mind that the majority of the trials looked at cycling (n=61) and running (n=28) performance, with only a limited number of studies examining knee extension (n=10) and rowing (n=5) performance. Considering that the effect sizes were similar for these exercise types, it's likely that the analyses for knee extension and rowing were simply underpowered to detect a significant effect. On the other hand, NO_3^- supplementation led to much larger improvements in handgrip exercise performance. This suggests that the ergogenic effects of NO_3^- may be greater in small muscle groups exercises, where performance is likely limited by peripheral factors rather than cardiac output. However, note that the handgrip subgroup analysis included only four trials.

The results from the subgroup analyses according to NO_3^- supplementation parameters suggest that daily doses of more than 5 mmol (310 milligrams) of NO_3^- are required to elicit a potent ergogenic effect, that even one day of supplementation is enough to improve training performance, and that NO_3^- should be taken 2–3.5 hours before training. These results are in line with the recommendations of [previous research](#)^[13].

One strength of this meta-analysis is the use of a random effects model to analyze the data, as this shows that the researchers didn't assume that the trials employed similar methodologies. The random effects model also tends to result in wider confidence intervals than a fixed effects model, and is more conservative in detecting significant results, which means that the researchers can be more confident that the detected effects are real.

This meta-analysis has some noteworthy limitations. First, although a risk of bias assessment was performed, it was not fully presented. As such, it's unclear if the results of the meta-analysis are based on high quality evidence. Second, heterogeneity was not assessed (or at least reported) for any of the analyses. Third, some of the subgroups (e.g., the women-only subgroup, and the handgrip, knee extension, and rowing subgroups) included a small number of trials, which means that they may have been underpowered to detect significant effects.

Finally, it's important to consider the generalizability of the results. The studies included in the meta-analysis included young, healthy, adult participants, which means that the results can't be generalized to older populations or to people with health conditions.

NO_3^- supplementation has a significant, but very small, benefit for training performance under both normoxia and hypoxia in young, healthy adults who are sedentary or recreationally trained, but not in highly trained athletes. These ergogenic effects may become

smaller with longer exercise durations. While the results suggest that the ergogenicity of NO_3^- supplementation may be blunted in women, and in knee extension and rowing exercises, more research is needed before definitive conclusions can be drawn.

Frequently asked questions

Q. How do I convert nitrate from mmol to milligrams?

One mmol corresponds to 62 milligrams of NO_3^- . This means that the ergogenic daily NO_3^- dose of 5–25 mmol per day corresponds to 310–1,550 milligrams, or approximately 200–1,000 grams of beetroot.

Q. Can I get an ergogenic dose of nitrates from my diet alone?

According to one study^[14], “normal” omnivorous diets supply approximately 74 milligrams of NO_3^- per day, while vegetarian diets supply around 267 milligrams per day. So, in short, yes, although you would have to increase your intake of specific vegetables that are especially high in NO_3^- .

The rough nitrate contents of various foods are shown in Figure 3.

Figure 3: The approximate nitrate content of different foods

Very low

(< 20mg/100g)

Asparagus
Eggplant
Onion
Green bean
Mushroom
Pea
Pepper
Potato
Tomato
Watermelon

Low

(20-50mg/100g)

Broccoli
Carrot
Cauliflower
Cucumber



Medium

(50-100mg/100g)

Cabbage
Dill
Turnip

High

(100-250mg/100g)

Chinese cabbage
Endive
Fennel
Leek
Parsley

Very high

(>250mg/100g)

Celery
Cress
Lettuce
Red beetroot
Spinach



Q. Are isolated nitrate supplements more or less effective than beet/vegetable extracts high in nitrate?

Almost 90% of the available trials looking at the ergogenicity of nitrates have used vegetable extracts (almost exclusively beetroot juice) as a form of NO_3^- supplementation. Out of the remaining 10% of the trials that have used isolated nitrate supplements (specifically, sodium or potassium nitrate) there seems to be^[10] roughly an equal split between trials finding a significant or non-significant ergogenic effect. This is largely in line with the results of the overall research on nitrate ergogenicity, and suggests that isolated nitrate supplements are similarly effective to vegetable extracts. However, more trials using isolated nitrate supplements are needed before any conclusions can be drawn with certainty.

What should I know?

There is substantial variability in the ergogenic effects of NO_3^- supplementation within and between trials, which is likely due to several individual, environmental, exercise-related, and supplementation-related factors. This meta-analysis examined the influence of these factors.

According to the results of this meta-analysis, NO_3^- supplementation in young, healthy adults has a very small effect on improving training performance that is unaffected by environmental oxygen levels. However, the improvements are not experienced by highly trained athletes, and the effects may be smaller with longer exercise durations. While the ergogenicity of NO_3^- supplementation may be blunted in women due to biological sex differences, and in knee extension and rowing exercises, only a handful of trials were included in these subgroups, so more research is needed to draw definitive conclusions.

Nitrate's ergogenic effects were optimized when participants took a daily dose of about 5–25mmol (310–1,550 milligrams) 2–3.5 hours before exercise after at least one day of prior supplementation.

What are your thoughts regarding nitrate supplementation? Do you have a good recipe for nitrate-containing vegetables? Share them in the [private NERD Facebook forum](#)!

[^ Go back to table of contents](#)

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Can sweet and salty mouth rinses provide a taste of victory?

🔖 Tags: Mouth Rinse, Exercise Performance, Endurance Training

📄 Study under review: [Effect of glucose and sodium chloride mouth rinses on neuromuscular fatigue: a preliminary study](#)

Quick Takes

- **What question this study addressed:** Can using sugar or salt mouth rinses while cycling attenuate reduction in muscle function?
- **Who this study applies to:** Young, healthy men.
- **What the intervention was:** Using a 6% sodium chloride or 6% glucose mouth rinse immediately before, and every 10 minutes during, a 30-minute cycling bout at 70% of maximal oxygen uptake.
- **What the main takeaway is:** Compared to a water rinse, both the salt and sugar mouth rinses mitigated the decline in muscle contraction after the cycling bout.
- **Caveats:** The study could not be blinded, was not preregistered, and had a very small sample size. Also, the primary outcome of muscular contraction is not a direct predictor of real-world sports performance. More studies are

needed in a wider variety of conditions to confirm this finding and to determine if it indicates actual benefits for performance.

What was the question?

The researchers designed the study to determine whether salt and carbohydrate mouth rinses could improve the perseverance of maximal voluntary contraction following a 30-minute cycling exercise at 70% of the participants' maximal oxygen uptake.

Why was the question worth asking?

A 2004 hallmark study^[1] found that using a carbohydrate mouth rinse solution could augment performance by stimulating receptors in the oral cavity. Since then, multiple studies have tried to pin down the exact mechanisms through which this effect could arise for both shorter^[3] and longer^[3] distance exercises.

Upon tasting sweetness, there is a gustatory excitation^[4] response in the body, during which receptors that reside in the oral cavity activate the parts of the brain^[5] that drive exertion^[6] during exercise. Scientists have proposed that the taste of sweetness exciting these areas leads to a response from these areas that counters the decline in potentially performance relevant neuro-muscular activation. In other words, merely tasting sweet things could excite parts of the nervous system, which could have ergogenic benefits. But carbohydrates aren't the only tastes that can stimulate regions of the brain that are important for movement—other tastes could do this, too^[7].

There are a lot of taste buds^[8] throughout the oral cavity, as well as in the gastrointestinal tract. Some respond to sweet sensations, others respond to salty or bitter ones. It's quite possible that stimulating these other taste receptors could provide a similar ergogenic effect as carbohydrate mouth rinses, through a similar mechanism. In fact, there's some evidence of this effect. For example, salt has been found to acutely produce an increase in muscle contraction force^[9] and can cause a neurobiological change in brain regions^[10] associated with [endurance exercise performance](#).

Since it is common practice for athletes to consume both carbohydrates and electrolytes^[11] (read: salts) during exercise, the authors of the study under review had a plausible reason to investigate

whether swirling a salt solution around the mouth and then spitting it out could have a positive (or detrimental) effect on athletic performance.

Carbohydrate mouth rinses stimulate areas of the brain that drive exertion during exercise, which could lead to ergogenic benefits. Other tastes, like salt, could have similar effects, in theory. The authors of the study under review aimed to test this hypothesis.

How was the question answered?

Ten active and young (average age of 22) males free of any metabolic diseases and/or injuries that would prevent them from completing the trials were recruited for the study. Participants were moderately fit, and had an above-average^[12] maximal oxygen uptake of 48.3 ml/kg/min.

Upon successful recruitment, the research team implemented a randomized cross-over design with a seven-day washout period between the three conditions: (1) water, (2) 6% glucose, or (3) 6% salt mouth rinse solutions. Participants were asked to refrain from exercise, caffeine, and alcohol consumption 24 hours prior to the trial.

Participants arrived at the lab for testing in a fasted state. They warmed up and then performed three sets of maximal isometric knee extensions, with a two-minute recovery between sets. The highest power produced from the three sets was used for the maximal voluntary contraction (MVC). An additional maximal isometric knee extension was performed for 90 seconds to assess sustained MVC (sMVC).

Once baseline measurements were taken, participants cycled for 30 minutes at 70% of their maximal oxygen uptake while rinsing their mouth out (five-second rinse) with one of the trial solutions in undisclosed amounts immediately before and every 10 minutes during the trial. The participants partook in MVC and sMVC testing once more before exiting the laboratory.

The study was not preregistered, no power calculations were performed, and no primary outcomes were declared. Moreover, there was no blinding of the treatments due to the fundamentally different tastes of the treatments.

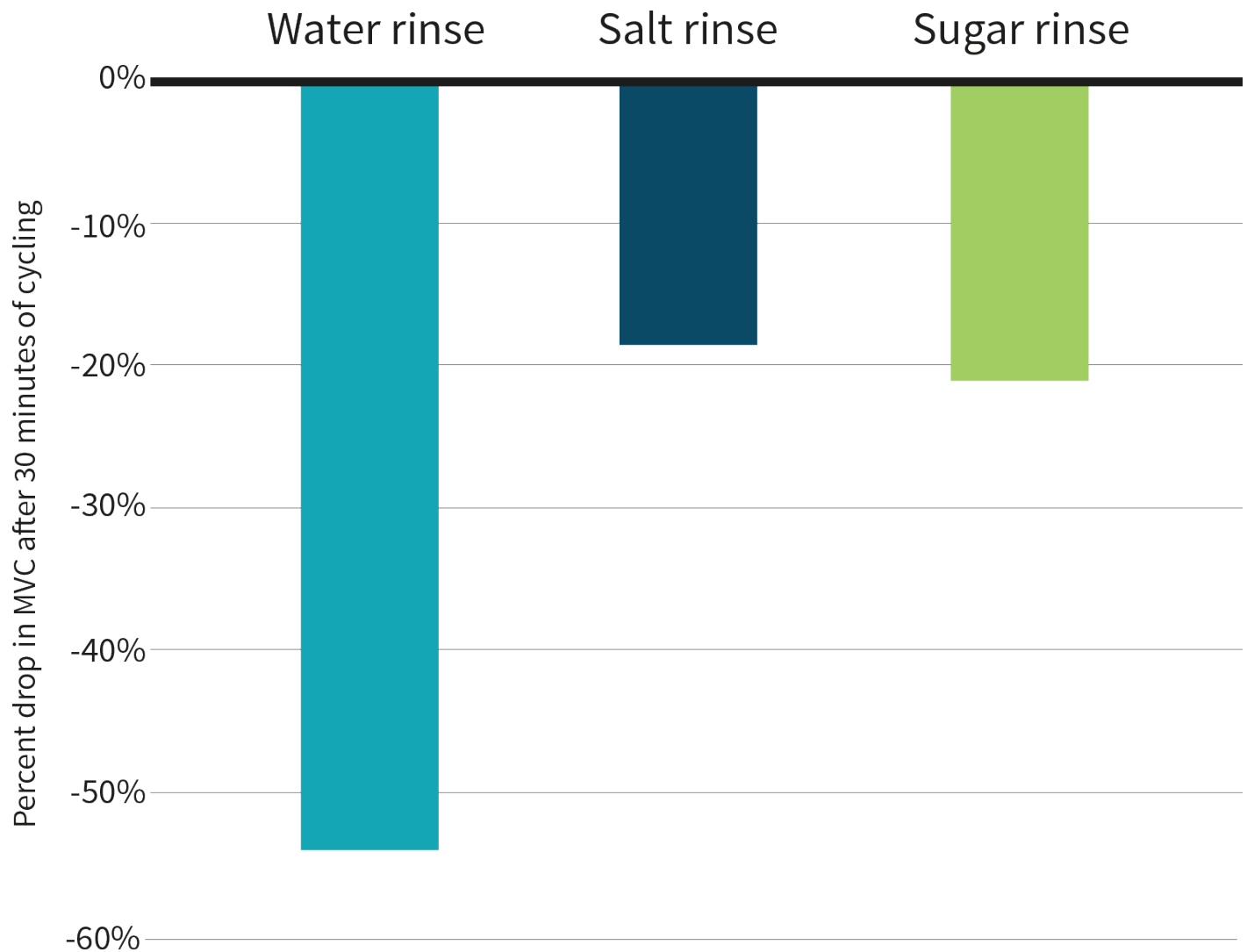
Changes in ten active young men's maximal and sustained knee

isometric contraction were measured following 30 minutes of cycling after swishing with either water, glucose solution, or salt solution every 10 minutes throughout the cycling exercise.

What was the answer?

Post-cycling testing revealed a significant reduction in MVC and sMVC for all conditions. However, glucose and salt mouth rinsing at 6% solutions preserved MVC and sMVC to a greater extent when compared to water, with no significant differences between glucose or salt mouth rinsing trials. Figure 1 shows that MVC dropped roughly 15% in the glucose and salt conditions, compared to a 35% drop when swishing water.

Figure 1: Percent drop in MVC in the three conditions



How much should you trust the answer?

The results of the study suggest that, in active and fasted men, a 6% carbohydrate or salt solution provided before and every 10 minutes during exercise preserves the participants' ability to produce force and sustained muscular contraction when compared to a water beverage.

A large portion of the research supports these findings^[13], as well as the idea that fasting results in taste receptors in the mouth and gut being more sensitive, thus sending a greater putatively ergogenic stimulus to the brain regions. Carbohydrate mouth rinsing, in particular, has been shown to improve performance in untrained^[14], trained^[15], endurance^[3], and strength training^[16] scenarios. However, there is also a great deal of literature suggesting carbohydrate mouth rinsing

doesn't work. Many of the studies that provided these results were done either in a fed state^[17], in the heat^[18], during repeated sprints^[19], during time trials in trained cyclists^[17] or included participants who were and remained euhydrated^[20].

It shouldn't go unmentioned that shoddy statistical analysis that has plagued the world of sport science^[21] could be to blame for overestimating positive findings in highly trained athletes partaking in MVC efforts^[16] in a fatigued state or during prolonged submaximal cycling exercise^[22]. Or if ingesting a carbohydrate beverage^[23] would elicit a better performance-enhancing effect than swishing and spitting.

In view of the heterogeneous study results of sweet mouth rinses, it is all the more warranted to narrow the focus of the discussion to the effects of rinsing with essentially salt water. As mentioned earlier, salt taste receptors are found throughout the oral cavity, influence insular cortex and/or opioid receptor activity^[24], and in turn, directly or indirectly affect autonomic activity^[25] and fatigue susceptibility^[26].

From that, it's reasonable to infer that rinsing your mouth with common electrolyte beverages acts as a way to minimize fatigue and preserve your ability to produce force over time. In fact, this latest addition to the 'mouth rinsing for exercise performance' research doesn't only suggest that salt works, but also that it is as effective as glucose when it comes to the amelioration of exercise-induced reductions in muscular contractility. However, follow-up studies are greatly needed before any conclusions can be stated with confidence, especially given the relatively small sample size and highly specific study population of young fit men in the study at hand. In addition to sample size and population considerations, there are some other issues regarding the study design that may limit placing too much confidence in its findings.

First, the participants couldn't be blinded to the intervention because salt tastes different than sweetness, leaving open the possibility that participants could have been primed to perform differently based on their previous beliefs about tastes. Researchers have noted that placebos have neurobiological effects^[27] and a recent review highlights the performance-enhancing effects of rinsing one's mouth out with a carbohydrate solution could be, in part, explained by a placebo (or nocebo) effect^[28].

Second, MVC isn't a direct measure of what most athletes actually care about: performance. So, it's open to question whether these results would have practical effects in terms of real-world performance benefits, such as improved time to exhaustion, better time-trial performance, etc.

Third, the authors could have benefited from an additional, fourth arm to the study, in which they combined the carbohydrate and salt sources into one. This would have been useful because it is common practice for athletes to consume both carbohydrates and electrolytes^[11] during exercise. Also, there's some physiological reasons to suspect a possible synergy between salty and sweet

tastes, such as the existence of a sodium-dependent glucose transporter^[29] (SGLT1) present in taste receptors, which requires taking up sodium along with glucose.

This study replicated similar effects found in studies involving carbohydrate mouth rinses while extending them to salt rinses also. However, these results should be considered preliminary, and ideally should be replicated in larger, more diverse participant groups and in studies that introduce a no-rinse condition or combined treatment with both carbs and salt.

What's the take-home?

This study confirmed that rinsing with a 6% carbohydrate solution has ergogenic effects on one's ability to produce force after a fatiguing bout of exercise in a fasted state. In addition, it found that similar benefits can also be had by rinsing with a 6% salt solution. More research in larger, more diverse populations that focus specifically on sports performance outcomes would be useful to confirm this effect and determine whether it has any practical benefits in a wider range of athletes.

Think these results are too sweet to be believed? Get salty over the evidence at the [NERD Facebook forum](#).

[^ Go back to table of contents](#)

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Not just for making cakes: baking soda may improve exercise performance

Tags: Sodium Bicarbonate, Endurance Training, Muscle Protein Synthesis

Study under review: Effects of Sodium Bicarbonate Supplementation on Muscular Strength and Endurance: A Systematic Review and Meta-analysis

Quick Takes

- **What question this study addressed:** Does supplementing sodium bicarbonate before exercise acutely improve muscular strength or endurance?
- **How the question was answered:** Through meta-analyses of randomized controlled trials.
- **Who was studied:** Almost exclusively males, ranging in fitness levels from athletes to untrained individuals.
- **What the intervention was:** Almost all trials used 0.3 grams of sodium bicarbonate per kilogram of bodyweight one to three hours before exercise.
- **What the main takeaway is:** Supplementing before exercise led to small improvements in muscular endurance, but there was no clear effect on muscular strength.
- **Caveats:** The total and individual number of participants was very low, leaving room for more research, especially in

women. The evidence to date is consistent with effect sizes ranging from trivial to moderate, and may very well vary depending on training status and possibly muscle group, given the differences in participants and muscles tested between trials.

What was the question?

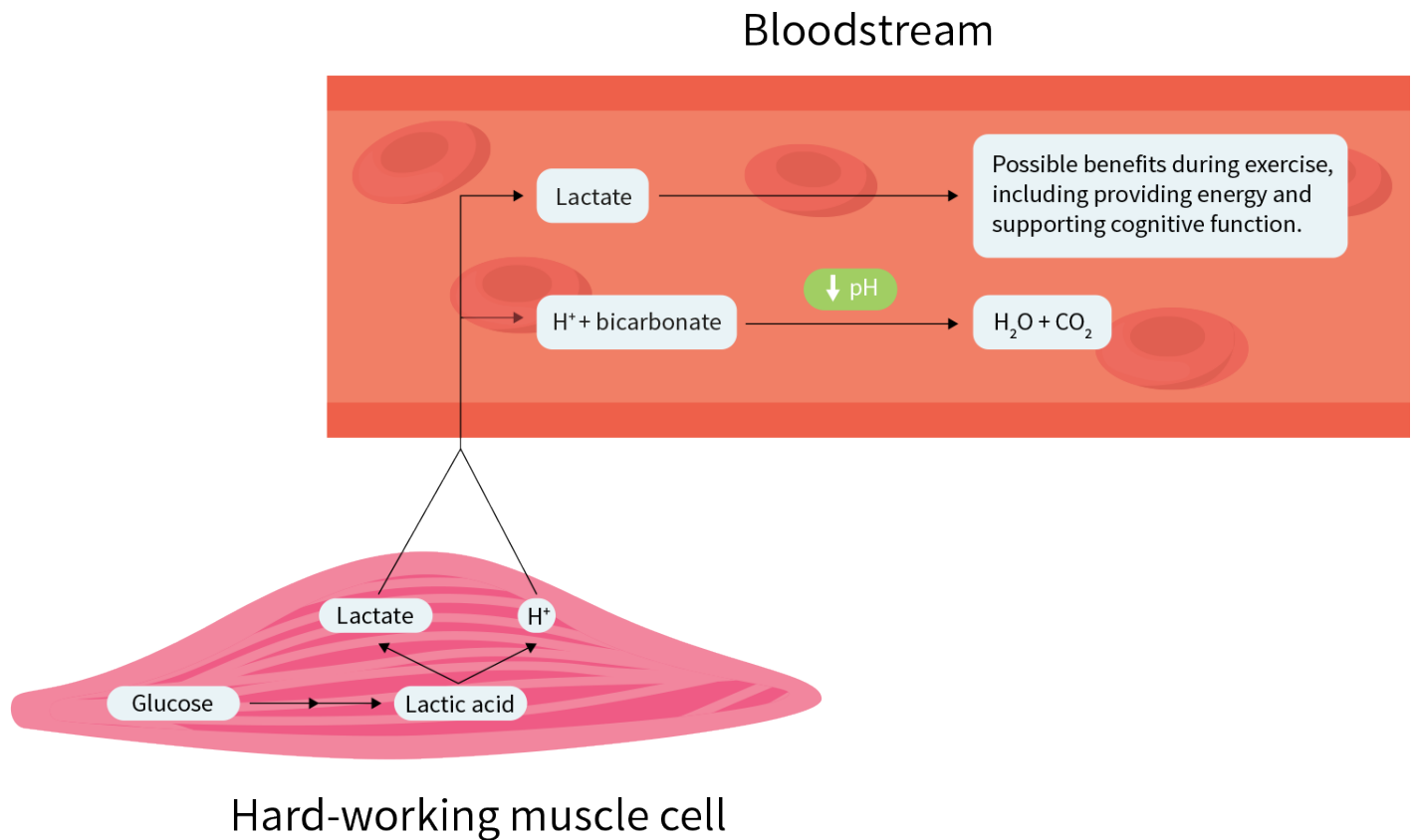
This meta-analysis was designed to determine whether supplementation with sodium bicarbonate (NaHCO_3) before exercise can improve muscular strength and endurance.

Why was the question worth asking?

Sodium bicarbonate, also known as baking soda or its chemical nomenclature NaHCO_3 , is a common household item used for a variety of purposes, including baking, pest control, fire extinguishing, cleaning, and personal hygiene. In 2018, it was included in the International Olympic Committee consensus statement^[1] as a supplement with good evidence for improving exercise performance.

Sodium bicarbonate is thought to improve exercise performance by increasing the concentration of blood bicarbonate, a buffer that can help maintain acid-base balance^[2]. Specifically, muscle fatigue during exercise is thought to take place, in part, due to exercise-induced acidosis^[3] (i.e. a decrease in pH). This means that the onset of muscle fatigue is partly dependent on the body's ability to buffer decreases in pH. Buffering agents like sodium bicarbonate^[4] neutralize hydrogen ions (H^+), helping to delay decreases in pH, thereby delaying muscle fatigue and improving exercise performance. This process is depicted in Figure 1.

Figure 1: How bicarbonate buffers acidity during exercise



Reference: Todd. *Biosci Horiz.* 2014 Jun. DOI: [10.1093/biohorizons/hzu001](https://doi.org/10.1093/biohorizons/hzu001)

While the general ergogenic effects of sodium bicarbonate and its physiological modes of action are relatively well-established, its practical benefits, specifically on muscular strength and endurance, are not. This is because the available trials that have investigated sodium bicarbonate's effects on muscular strength and endurance have produced conflicting results; and although several meta-analyses^{[5][6][7]} have explored the effects of sodium bicarbonate on various aspects of exercise performance (with predominantly positive results for sustained high-intensity exercise with high anaerobic metabolic demands), none have focused specifically on muscular strength and endurance.

The goal of the study under review was to meta-analyze the available trials in order to determine the overall impact of acute sodium bicarbonate supplementation on muscular strength and endurance.

The well-established ergogenic effects of sodium bicarbonate coupled with mechanistic evidence on its physiological modes of action suggest that acute sodium bicarbonate supplementation

may have positive effects on muscular strength and endurance. However, the available relevant trials have produced conflicting results, prompting the study authors to conduct this meta-analysis.

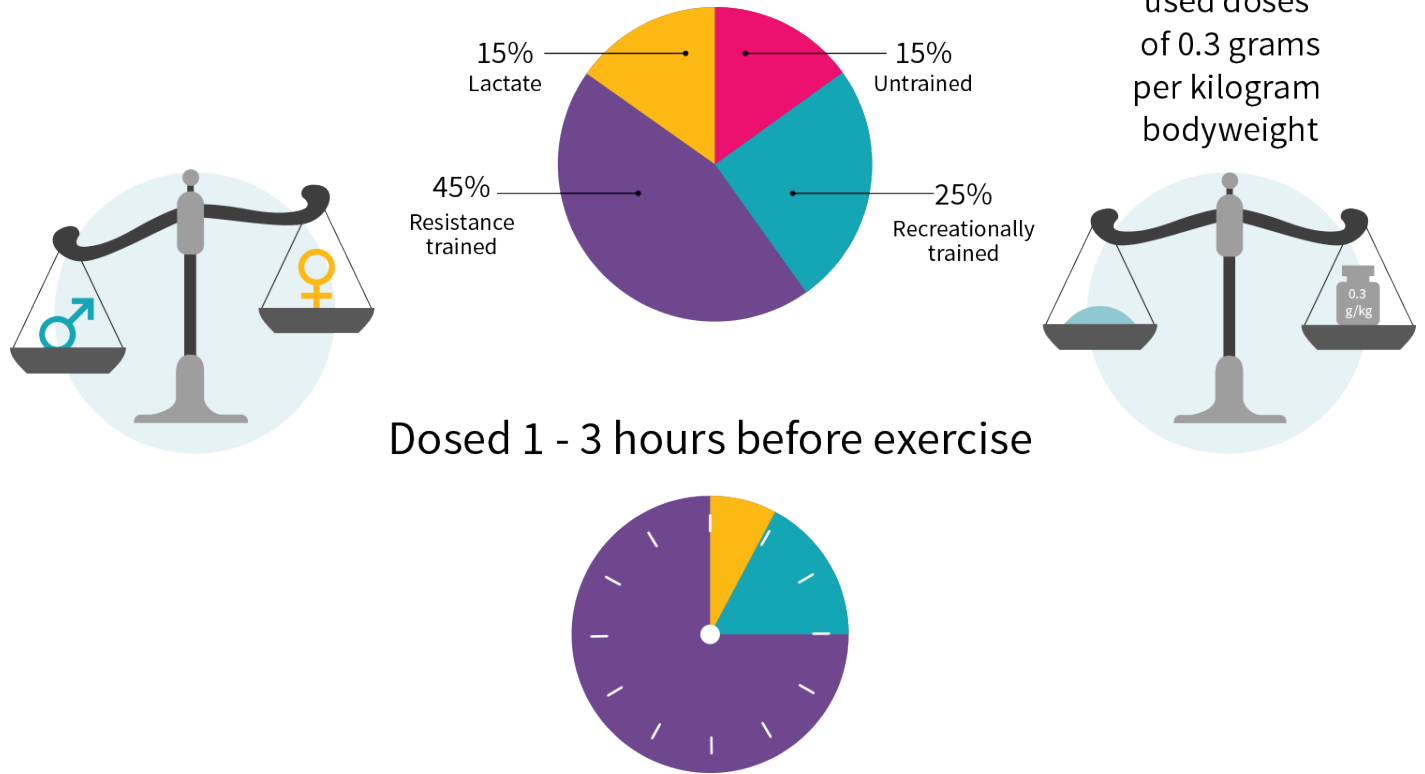
How was the question answered?

The authors tried to answer the question by conducting a systematic review and meta-analysis of crossover trials that looked at the effects of acute sodium bicarbonate supplementation on muscular strength and endurance in healthy individuals. In the trials included in the meta-analysis, muscular strength was assessed by changes in peak force, peak torque, or maximum load lifted, while muscular endurance was assessed by changes in the number of repetitions performed, isokinetic total work, or time to maintain isometric force production.

Ultimately, 20 trials were included in the meta-analysis, with 13 trials looking at muscular endurance, and 11 looking at muscular strength. The total number of participants across all trials was 192 (median number of trial participants was 10), with 184 men and eight women. As shown in Figure 2, the participants of the trials included untrained, recreationally trained, and resistance-trained individuals, as well as athletes. Out of the 20 trials included, 19 used a sodium bicarbonate dose of 0.3 grams per kilogram of bodyweight, and one used a dose of 0.4 grams per kilogram of bodyweight. The timing of ingestion ranged from 60 to 180 minutes before exercise.

Figure 2: Major study characteristics

Study populations



The muscular performance data of the included trials were converted to [standardized mean differences](#) (SMD) and summarized in a random-effects model. Subgroup analyses were performed for the muscular endurance of small vs. large muscle groups, and for muscular strength in a rested vs. fatigued state. Heterogeneity was tested using the I^2 statistic, while the methodological quality of the included trials was assessed with the Physiotherapy Evidence Database^[8] (PEDro) checklist. The study under review was conducted according to [PRISMA](#) guidelines, but does not appear to have been preregistered.

This was a systematic review and meta-analysis of 20 trials involving a total of 192 healthy participants (of which 184 were men) that explored the effects of supplementation with 0.3-0.4 grams of sodium bicarbonate per kilogram of bodyweight 60 to 180 minutes before exercise on muscular strength and endurance.

What was the answer?

The meta-analysis found that, in healthy individuals, acute sodium bicarbonate supplementation had a small positive effect on muscular endurance (SMD = 0.37; 95% CI: 0.15, 0.59) that decreased in size, but remained significant (SMD = 0.20; 95% CI: 0.05, 0.36), after researchers excluded one study^[9] that reported a very large effect. In the subgroup analyses for this outcome, sodium bicarbonate improved muscular endurance for both large (SMD = 0.40; 95% CI: 0.13, 0.66) and small (SMD = 0.31; 95% CI: 0.04, 0.59) muscle groups.

For muscular strength, there were no significant effects of sodium bicarbonate supplementation (SMD = -0.03; 95% CI: -0.18, 0.12) whether the testing was carried out in a rested state (SMD = 0.02; 95% CI: -0.09, 0.13) or in a fatigued state (SMD = -0.16; 95% CI: -0.59, 0.28).

Of the trials included in the meta-analysis, 15 were rated as being of excellent methodological quality, four trials were assessed as good, and one trial was assessed as fair. None of the studies were assessed as being of poor methodological quality. Moderate levels of heterogeneity ($I^2 = 70-73\%$) were detected between trials for the primary outcomes.

The findings of this meta-analysis indicate that sodium bicarbonate supplementation before exercise can improve muscular endurance in both large and small muscle groups. It's unlikely to improve muscular strength in either a rested or fatigued state, however.

How much should you trust the answer?

The observed positive effects of sodium bicarbonate on muscular endurance make sense from a physiological standpoint. This is because muscular endurance tasks can result^[10] in high production of H^+ , and sodium bicarbonate is thought to elicit its performance-enhancing effects by removing H^+ from the working muscle, helping to delay decreases in pH, and, consequently, delaying muscle fatigue.

That said, it should be noted that the size of the effect was small, especially after one study^[9] with a very large effect size was removed (SMD = 0.20). Moreover, the 95% confidence intervals, which range from 0.15 (a trivial effect) to 0.59 (a medium effect size), allow for a wide range of effect sizes individual trainees can expect.. Moreover, moderate levels of heterogeneity were detected,

which suggests that there may have been important differences between trials.

One example of a potentially relevant difference is the training status of the participants. Since training can increase^[11] one's general muscle buffer capacity, it could be the case that the size of the effect of sodium bicarbonate supplementation varied according to the participants' training status. However, this is speculative and a potential topic of investigation for future research. Supplementation side effects may also contribute towards greater heterogeneity. Some people experience gastrointestinal effects^[12] from sodium bicarbonate supplementation, although the seriousness of these can vary greatly between individuals. Since gastric discomfort can negatively affect^[13] exercise performance, it's possible that greater ergogenic effects were found in trials where few or no participants experienced significant gastric discomfort, which may contribute toward increased heterogeneity.

The absence of a significant effect of sodium bicarbonate on muscular strength also makes sense from a physiological standpoint. Muscular strength tests typically involve maximal exertion of effort that only lasts a few seconds, while research^[1] suggests that the ergogenic effects of bicarbonate are seen during exhaustive exercise lasting between 30 seconds and 10 minutes, during which period anaerobic glycolysis^[14] plays a major role in energy provision. While it was hypothesized that ergogenic effects from sodium bicarbonate supplementation may have been more likely in a fatigued state due to possible H⁺ accumulation from previous exercise, the results of the meta-analysis did not detect such effects.

Some of the strengths of the study under review include the use of a random-effects model to analyze the data (which is more accurate in detecting significant results than a fixed-effect model), the inclusion of only crossover trials (which has each participant act as their own control and, therefore, minimizes inter-individual variance), and the use of mostly trials of good or excellent methodological quality.

However, the meta-analysis under review also has some limitations. First, it only included a relatively small number of trials, most of which had small samples sizes. Second, there were moderate levels of heterogeneity between the included trials, which suggests that it may not have been appropriate to meta-analyze the data from these trials to begin with. Finally, out of the 192 total participants, 184 were men, which means that the results cannot necessarily be extrapolated to women.

While the results of the study under review make sense from a physiological standpoint, the inclusion of a limited number of small trials with moderate levels of heterogeneity is an issue.

What's the take-home?

This meta-analysis suggests that, in healthy men, supplementation with 0.3-0.4 grams of sodium bicarbonate per kilogram of bodyweight 60 to 180 minutes before exercise will likely have a small positive effect on muscular endurance in both large and small muscle groups, but not on muscular strength, either in a rested or fatigued state. Larger trials are needed to confirm these findings, and more research is needed to investigate potential variations in efficacy in different populations.

Do you use baking soda for making cakes or making gains? Let us know in the private [NERD Facebook group](#)!

[^ Go back to table of contents](#)

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Supercharging muscle protein synthesis through essential amino acids

🔖 Tags: [EAAs](#), [Muscle Protein Synthesis](#), [Whey](#), [Muscle Mass](#)

📄 Study under review: [Anabolic response to essential amino acid plus whey protein composition is greater than whey protein alone in young healthy adults.](#)

Quick Takes

- **What question this study addressed:** How does a blend of whey protein and essential amino acids (EAAs) at two doses affect muscle anabolism compared to whey alone?
- **Who this study applies to:** Young, healthy men and women.
- **What the intervention was:** One or two doses of a commercial blend containing 3.2 grams of EAAs and 2.4 grams of protein per dose, compared to a commercial drink containing 12.6 grams of whey.
- **What the main takeaway is:** All groups experienced a boost in protein synthesis, with the high-dose EAA/protein blend seeing the highest.
- **Caveats:** Along with some open questions concerning study design and reporting, doses of EAAs, leucine, and whey protein in this study were all suboptimal. This is possibly due to the fact that this study tested commercial blends with set

ratios of these nutrients. Also, how the results found here relate to practical factors like lean mass and strength gains remains to be explored.

What was the question?

What is the protein synthesis response to a mixture of essential amino acids plus whey and a commercial drink (Gatorade Recover) containing 12.6 grams of whey?

Why was the question worth asking?

Changes in muscle mass are driven by changes in muscle protein synthesis^[1] (MPS). New protein is constantly being created while older proteins are degraded. This is called protein turnover. To build muscle, the rate of muscle protein synthesis must exceed the rate of muscle protein breakdown. Muscle protein synthesis can be increased by eating protein or moving. However, there are several factors that determine how much MPS responds, including age, exercise, and nutrition. There are lots of reasons to want to maximize MPS, from offsetting muscle loss due to diseases such as cancer or diabetes,^[2] to enhancing the hypertrophic response to training. Therefore, it could be useful for a wide range of people if researchers could identify a specific ingredient, dose, or mixture for maximizing MPS.

Protein synthesis can be measured using multiple methods^[3]. One is through measuring whole-body protein synthesis. Another involves measuring directly in the muscle, which is known as muscle protein fractional synthesis. These two measurements mostly agree in direction with each other, but the magnitude of change can be different. This is likely because muscle accounts for about 30% of whole-body protein synthesis and turns over at a much slower rate than other organs^[4], like the gut or liver. The study under review used both methods, which helps determine the effects of the different protein mixtures in relation to older research that only used whole-body protein synthesis.

Previous research has found that the post-meal rise in plasma essential amino acids^[5] (EAA) or leucine^[6] is important for the increase in MPS. It is currently recommended that 20-25 grams^[7] (8-10 grams of EAA) of a rapidly-absorbed protein, like whey^[8], can maximally stimulate MPS after resistance exercise in young healthy individuals. However, whether adding EAA to the mix can help

even further is still unclear. This study helps fill that gap by addressing the possible superiority of untested mixtures of EAA plus whey.

The authors argue that the amino acid with the lowest concentration relative to demand could limit the body's anabolic response, even if other amino acids dosages are high. This is why combining a mixture of all EAAs with an intact protein could be beneficial. It is also well known that the protein synthetic response to whey is longer in duration than free EAA^[9], but the spike in MPS with free-form essential amino acids is generally larger. So, if the two are combined there might be a further benefit for increasing protein synthesis.

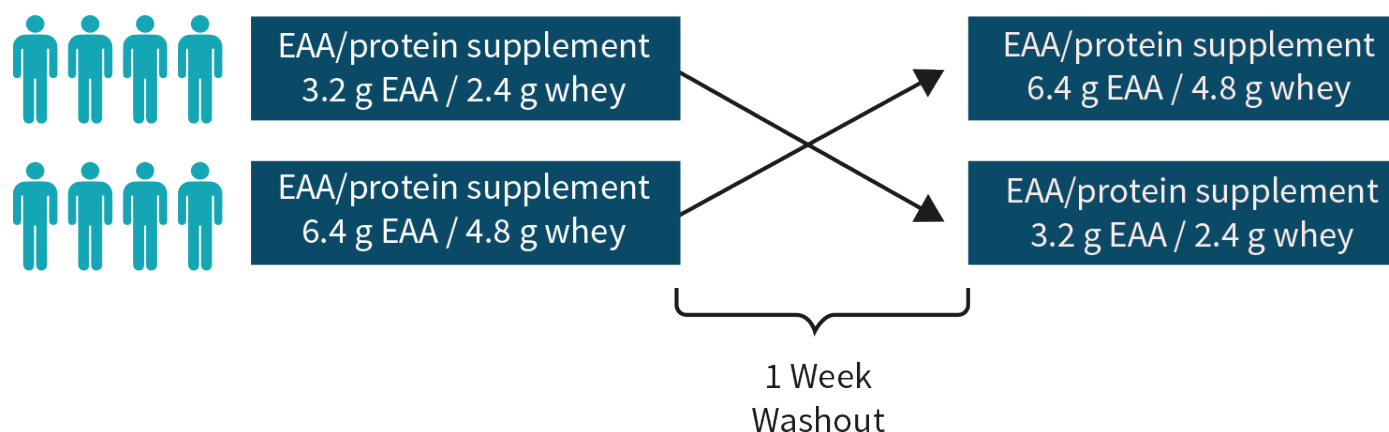
There is reason to suspect that adding essential amino acids (EAAs) to whey protein could improve muscle protein synthesis. However, optimal dosing is still unclear. This study explored how different doses of EAAs added on top of whey protein affects muscle protein synthesis.

How was the question answered?

This study used a randomized, two-period, stable isotope infusion in fasted and post-meal conditions. One group was randomized to drink a serving of an EAA and whey supplement mixture consisting of 3.2 grams of EAA and 2.4 grams of whey, or two servings of the same supplement, which equates to 6.4 grams of EAA and 4.8 grams of whey, then crossed over to take the other dose, as shown in Figure 1. In a separate arm, different participants were given 17.6 grams of Gatorade Recover, which contained 12.6 grams of whey protein.

Figure 1: Study design

ARM 1



ARM 2



The primary outcome was the total anabolic response, which is measured by subtracting muscle protein breakdown from muscle protein synthesis. The secondary outcomes were muscle protein fractional synthetic rate and plasma amino acid concentration. The study involved 16 college-aged female and male participants. The protocol was [preregistered](#) with its prespecified primary outcomes being whole-body protein balance and changes in muscle protein synthesis.

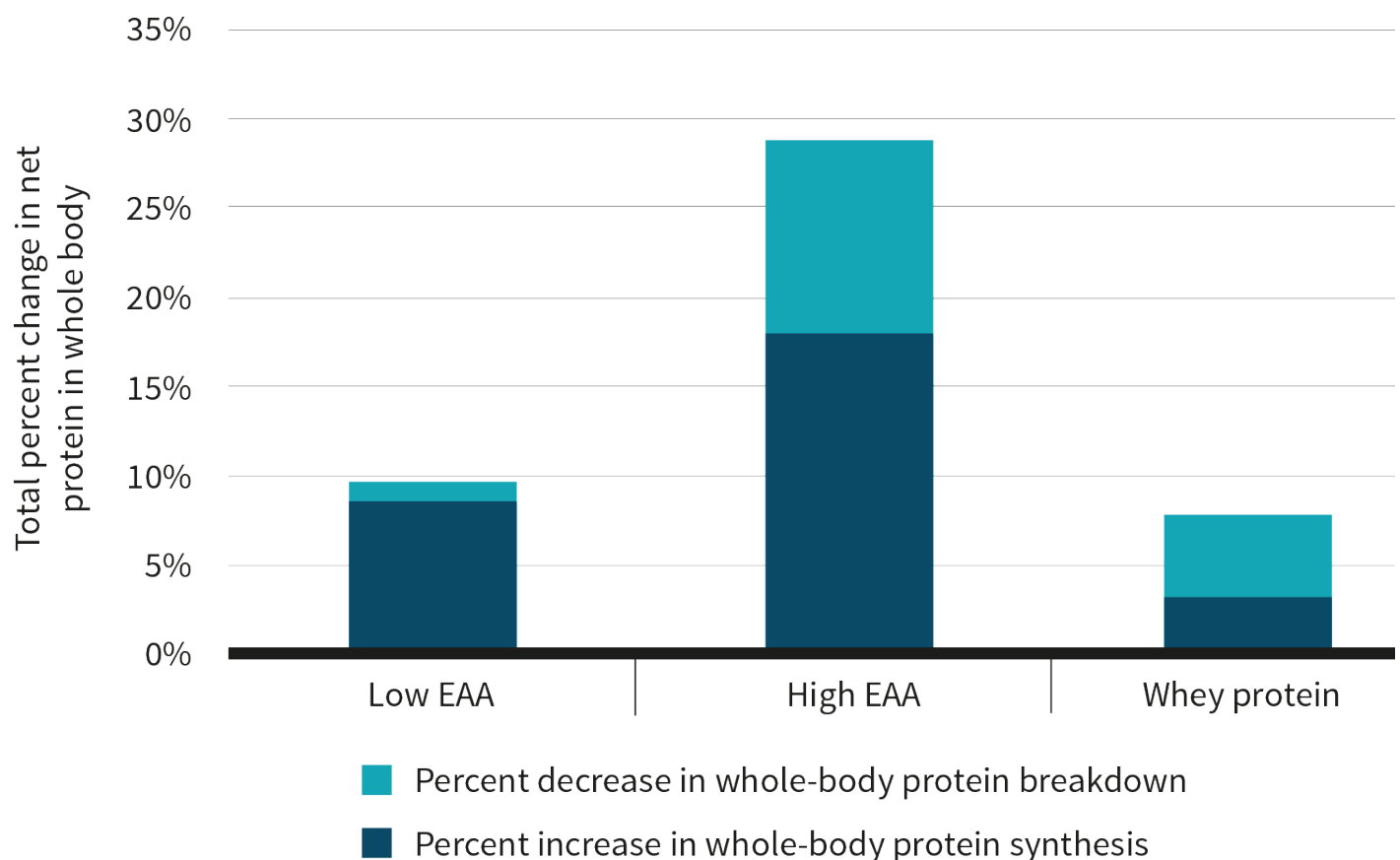
In this study, 16 college-aged participants were provided either a commercial drink containing 12.6 grams of whey protein, or a mix of EAA and whey containing 3.2 grams of EAA and 2.4 grams of whey, or 6.4 grams of EAA and 4.8 grams of whey. Muscle protein synthesis and whole-body protein balance were the primary

outcomes.

What was the answer?

Every group increased whole-body protein balance. The group consuming 12.6 grams of the EAA and whey mixture experienced a significant increase, while the results of the 6.3 gram EAA/whey group were roughly similar to the Gatorade Recover group. The total net changes in protein balance and the relative contributions to increasing protein synthesis and decreasing protein breakdown are shown in Figure 2.

Figure 2: Net-body whole protein changes



For muscle protein synthesis, only the 12.6 gram EAA/whey group significantly increased from baseline, while the 6.3 gram EAA/whey group and the Gatorade Recover group did not.

All conditions increased whole-body protein balance, with the higher dose of the EAA and whey combo having the biggest effect. Only the high-dose combo had a statistically significant effect on muscle protein synthesis.

How much should you trust the answer?

This study adds to the body of literature on protein synthesis because it used two doses of essential amino acids with whey that have not been tested directly, and compared them to a commercial whey protein product, Gatorade Recover. The authors found that a balanced, 12.6 gram mixture of essential amino acids with whey resulted in an improvement in net protein balance and fractional synthesis rate greater than that of a mixture of 6.3 grams of EAA with whey or Gatorade Recover.

Essential amino acids can increase MPS at rest^[10] and after strength training^[11], while nonessential amino acids do not. Furthermore, a dose-response study found the optimal essential amino acid dose to maximally stimulate muscle protein fractional synthesis rate is about 10 grams^[12], which somewhat agrees with the data presented in this study because a 12.6 gram mixture of EAA/whey, which contained 6.4 grams of free EAA, was the only treatment that had a statistically significant effect on MPS. However, the EAA mixture used here is well short of the optimal dose, so there could have been an even better MPS response with a higher EAA content. This raises the question of why the authors did not use a larger dose. One possible reason could be because they are comparing commercial products, including the EAA product of their study's sponsor, Adesso LLC.

The submaximal EAA dose isn't the only reason why there may be room for improvement in the tested proprietary blend. It is thought that leucine serves as a trigger for the MPS response. However, the high dose EAA/whey mixture only contained 128 milligrams of leucine, which is well below the proposed threshold of three grams of leucine^[13] for a maximal MPS response. Using a somewhat similar experimental design, another study found a dose of 15 grams of EAA with about 2.8 grams of leucine increased MPS by about 60%^[14], which is actually less than the increase reported here. However, an exact comparison is difficult because study participants had a higher baseline MPS due to some differences in experimental design.

The anabolic response to protein is frequently thought of as a muscle response, but whole body response is also important to compare to earlier research in the field. Furthermore, the whole body

approach has been used extensively in protein research because it is less invasive and can measure both rates of protein synthesis and breakdown. Here, the authors found that the higher dose EAA plus whey mixture had a higher net balance due to an increase in whole-body protein synthesis and a reduction in muscle protein breakdown. This data falls in line with the literature, as the comparison of whole body protein synthesis and muscle protein synthesis can be different^[15]. It also indicates that consuming EAA isn't solely elevating protein synthesis in the muscle because other tissues can use them too.

There is some debate about whether MPS reflects long term muscle growth. One study from 2014 found that MPS is not correlated with strength training^[16], which is known to increase muscle mass. However, another study found that MPS does correlate with muscle growth^[17]. Even more recently, a different group of researchers found muscle growth to correlate with an increase in MPS after three weeks of strength training^[18]. These studies reveal more about MPS post-exercise than with EAA intake. In fact, there have been no MPS studies that compare EAA or whey to long term muscle growth. This leaves a difficult to answer question: would these acute MPS elevations translate to gaining muscle mass? It's hard to say without longer-term data looking specifically at muscle mass.

It is worth mentioning that the muscle specific protein synthesis methods are limited to some extent by the assumption that the responses at a single site may not reflect all skeletal muscle^[19]. The study also does not indicate whether these participants were trained or untrained, which could play a role in how muscle protein fractional synthesis rates change^[20] over the course of the measuring period. Another issue is that while the methods used in the study match those in the preregistration, the researchers reported 24 participants in the trial registration. It's unclear why this discrepancy exists, but one possibility is that the authors counted the participants in the crossover trial twice in the preregistration. In addition, the pre-planned participant number was 30 and the authors didn't reach it, and no explanation was given. It's possible that the study was stopped short because significant differences were found. A power analysis could rule out this possibility, but if the authors did one, it wasn't reported. Furthermore, the comparison to Gatorade Recover is odd because one serving contains only 12.6 grams of whey. This is lower than other ready-to-drink shakes or a standard scoop of whey powdered protein and may make the EAA/whey mixture look better than if it were compared to the recommended 25-30 gram^[21] dose.

While this study found that the higher-dose EAA/whey blend improved muscle protein synthesis, the dose used may still be suboptimal, and was tested as part of a proprietary blend. Also, the control of pure whey used a lower-than-optimal dose as well,

raising questions about how well the tested blend would compare to a higher dose of whey. Finally, whether the findings would translate to long term muscle growth without strength training is an open question.

What's the take-home?

The authors found that a 12.6 gram mixture of EAA and whey increases muscle protein fractional synthesis beyond that of a 6.3 gram EAA/whey mixture or 17.6 grams of Gatorade Recover, which contains 12.6 grams of whey per serving. There were also changes in whole-body protein balance in all the groups, but the high dose EAA/whey group had the highest.

However, the amount of EAA in the high-dose blend group was well below the optimal amount of EAA previously found to maximize muscle protein fractional synthesis, while the dose of whey it was compared to was also suboptimal. These issues, along with some questions about study design and reporting, suggest that these results should be taken as preliminary.

Stop by the [NERD Facebook forum](#) to add to our non-proprietary blend of discussion around EAAs and gains!

[^ Go back to table of contents](#)

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